TODAY'S TOPICS 16/10/26

- RADIAL DISTRIBUTION FUNCTION (RDF) $g(r)$
- CALCULATION OF $g(r)$ FROM SIMULATIONS

RADIAL DISTRIBUTION FUNCTION

$\tilde{\rho}(\vec{r}_1, \vec{r}_2)$ = { PROBABILITY OF FINDING A PARTICLE IN $\vec{r}_1$ AND ANOTHER ONE IN $\vec{r}_2$ }

EVENT A = { A PARTICLE IS IN $\vec{r}_1$ }

EVENT B = { // // // // $\vec{r}_2$ }

$$P(A \cap B) = \tilde{\rho}(\vec{r}_1, \vec{r}_2)$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

↑ CONDITIONAL PROBABILITY

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

NOTE THAT (HOMOGENEOUS CASE):
IF $P(A) = P(B)$
$P(A|B) = \frac{P(A \cap B)}{P(B)} =$
$= \frac{P(A \cap B)}{P(A)} = P(B|A)$

IF A AND B INDEPENDENT

$$P(A|B) = \frac{P(A)\,\cancel{P(B)}}{\cancel{P(B)}} = P(A)$$

GOING BACK TO $\tilde{\rho}$:

$$\tilde{\rho}(\vec{r}_1, \vec{r}_2) = \tilde{\rho}(\vec{r}_1|\vec{r}_2)\,\tilde{\rho}(\vec{r}_2) \qquad (*)$$

WE DEFINE:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) \equiv \frac{\tilde{\rho}(\vec{r}_1, \vec{r}_2)}{\tilde{\rho}(\vec{r}_1)\,\tilde{\rho}_2(\vec{r}_2)}$$

} THIS IS 2-PARTICLE (PAIR) DISTRIBUTION FUNCTION

IF THE SYSTEM IS HOMOGENEOUS $\tilde{\rho}(\vec{r}_1) = \tilde{\rho}(\vec{r}_2) = \tilde{\rho}$

SO THAT:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}(\vec{r}_1, \vec{r}_2)}{\tilde{\rho}^2}$$

BUT USING (*) WE HAVE:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}(\vec{r}_1|\vec{r}_2)\,\cancel{\tilde{\rho}}}{\tilde{\rho}\,\cancel{\tilde{\rho}}}$$

$$= \frac{\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2)}{\tilde{\rho}}$$

SINCE SYSTEM IS HOMOGENEOUS WE CAN EXPECT

$g^{(2)}(\vec{r}_1, \vec{r}_2) = g^{(2)}(\vec{r}_1 - \vec{r}_2)$

SO THAT $g^{(2)}(\vec{r}_1 - \vec{r}_2) = g^{(2)}(\vec{r})$

AND $\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2) = \tilde{\rho}(\vec{r})$

I.E. $\tilde{\rho}\, g^{(2)}(\vec{r}_1, \vec{r}_2) = \tilde{\rho}(\vec{r}_1 \mid \vec{r}_2)$

IDEAL GAS

$$\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2) = \tilde{\rho}(\vec{r}_1) = \tilde{\rho}$$

HENCE

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}}{\tilde{\rho}} = 1$$

HARD SPHERES

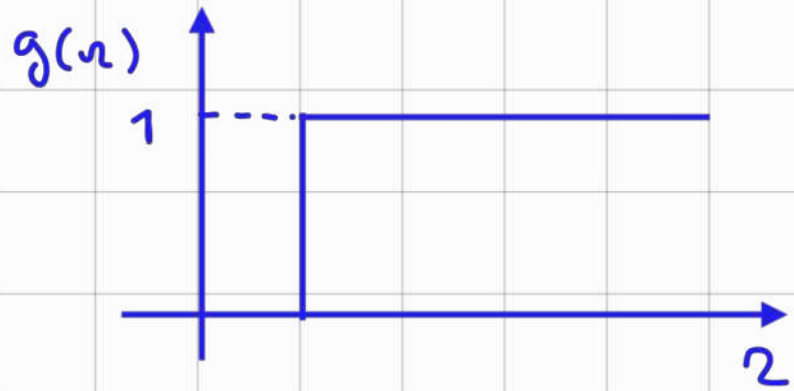
IF $\|\vec{r}_1 - \vec{r}_2\| < \sigma \Rightarrow \rho^{(2)}(\vec{r}_1 \mid \vec{r}_2) = 0$

AND $g^{(2)}(\vec{r}_1 \mid \vec{r}_2) = 0$

MOREOVER SYSTEM IS ISOTROPIC THEREFORE

$g^{(2)}(\vec{r}_1 \mid \vec{r}_2) = g^{(2)}(\|\vec{r}_{12}\|) = g^{(2)}(r)$

RADIAL DISTRIBUTION FUNCTION



AT VERY LOW DENSITIES

BUT AT HIGH DENSITIES ?

WE WILL SEE ...

TWO POINTS DENSITY

IF $\mathcal{N}(\vec{r}_1 \dots \vec{r}_N) = \dfrac{e^{-\beta U_N}}{Z}$

$\mathcal{N}(\vec{r}_1 \dots \vec{r}_N)\, d\vec{r}_1 \dots d\vec{r}_N$ IS THE PROBABILITY OF FINDING PARTICLES AT POSITIONS $\vec{r}_1, \dots \vec{r}_N$

HENCE THE PROBABILITY DENSITY TO FIND ONE PARTICLE IN $\vec{r}_1$ IS:

$$\tilde{\rho}(\vec{r}_1) = \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_2 \dots d\vec{r}_N$$

NOTE THAT WE ARE NOT INTEGRATING OVER $d\vec{r}_1$

AND THE (1-PARTICLE) DENSITY IS:

$$\rho(\vec{r}_1) = N \cdot \tilde{\rho}(\vec{r}_1) = N \cdot \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_2 \dots d\vec{r}_N$$

N: TOTAL NUMBER OF PARTICLES

NOTE THAT $\int \tilde{\rho}(\vec{r}_1)\, d\vec{r}_1 = 1$ AND THAT $N \tilde{\rho}(\vec{r}_1)\, d\vec{r} = \{$ NUMBER OF PARTICES IN $d\vec{r}_1 \}$

MOREOVER IF THE SYSTEM IS HOMOGENEOUS

$$\rho(\vec{r}_1) = \frac{N}{V}$$

SIMILARLY THE PROBABILITY DENSITY OF FINDING A PARTICLE IN $\vec{r}_1$ AND $\vec{r}_2$ IS

$$\tilde{\rho}(\vec{r}_1, \vec{r}_2) = \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_3 \dots d\vec{r}_N$$

NOTE THAT WE ARE NOT INTEGRATING OVER $\vec{r}_1$ AND $\vec{r}_2$

SO THAT

$\underbrace{N(N-1)}_{\downarrow}\ \tilde{\rho}(\vec{r}_1,\vec{r}_2) = \{$ NUMBER OF PAIRS IN $d\vec{r}_1$ AND $d\vec{r}_2$ $\}$

TOTAL NUMBER OF POSSIBLE PAIRS

AND WE DEFINE THE TWO PARTICLE DENSITY AS FOLLOWS

$$\rho^{(2)}(\vec{r}_1,\vec{r}_2) = N(N-1)\int \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r}$$

IDEAL GAS

$$\rho^{(2)}(\vec{r}_1,\vec{r}_2) = N(N-1)\int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_3 \ldots d\vec{r}_N$$

$U_N = 0$

$$= N(N-1)\int \frac{d\vec{r}_3 \ldots d\vec{r}_N}{Z} = \overbrace{N(N-1)}^{\approx N^2}\frac{V^{N-2}}{V^N} =$$

SINCE $Z = \int e^{-\beta U_N} d\vec{r}_1 \ldots d\vec{r}_N = V^N$

$$\approx \frac{N^2}{V^2} = \rho^2 \quad \text{HENCE}$$

$$g^{(2)}(\vec{r}_1,\vec{r}_2) = \frac{\rho^{(2)}(\vec{r}_1,\vec{r}_2)}{\rho^2} = \frac{\rho^2}{\rho^2} = 1$$

ALTERNATIVE WAYS OF CALCULATING 1-POINT AND 2-POINT DENSITIES

$U_N = U_N(\vec{r}_1,\vec{r}_2 \ldots \vec{r}_N)$

$$\rho(\vec{r}) = \frac{N}{Z}\int e^{-\beta U_N} d^{(N-1)}\vec{r} =$$

NOW $U_N = U_N(\vec{r}_1 \ldots \vec{r}_N)$

$$= \frac{N}{Z}\int e^{-\beta U_N}\, \delta(\vec{r}-\vec{r}_1)\, d^N\vec{r} =$$

INTEGRAL DOES NOT DEPEND ON PARTICLE INDEX

$$= \frac{1}{N}\sum_i^N \int e^{-\beta U_N}\, \delta(\vec{r}-\vec{r}_i)\, d^N\vec{r}$$

TAKE A SYSTEM OF 3 PARTICLES CONSIDER $\vec{r}_2$ AND $\vec{r}_3$ INSTEAD OF $\vec{r}_1$, THEN SWAP INDICES TO RECOVER CASE ABOVE ($\vec{r}_1$)

$$= \left\langle \sum_i^N \delta(\vec{r}-\vec{r}_i) \right\rangle$$

NOTE THAT $U_2(\vec{r}_{iT}) = U_2(\vec{r}_{Ti})$

WHERE $\langle \dots \rangle = \int \dots \frac{e^{-\beta U_N}}{Z} d^N\vec{r}$

SIMILARLY

$$\rho^{(2)}(\vec{r}', \vec{r}) = \left\langle \sum_{i,j}{}' \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \right\rangle$$

($'$: $i \neq j$)

INDEED

$$\rho^{(2)}(\vec{r}', \vec{r}) = N(N-1) \int \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r} =$$

($U_N = U_N(\vec{r}', \vec{r}, \vec{r}_3 \dots \vec{r}_N)$; $U_N = U_N(\vec{r}_1 \dots \vec{r}_N)$)

$$= N(N-1) \int \delta(\vec{r}' - \vec{r}_1)\, \delta(\vec{r} - \vec{r}_2) \frac{e^{-\beta U_N}}{Z} d\vec{r}_1 d\vec{r}_2 \cdot d^{(N-2)}\vec{r}$$

TAKE AGAIN A SYSTEM OF 3 PARTICLES AND USE $(\vec{r}_2, \vec{r}_3)$ INSTEAD OF $(\vec{r}_1, \vec{r}_2)$

INTEGRAL DOES NOT DEPEND ON WHICH PAIR OF INDICES WE ARE CONSIDERING

$$= \sum_{i,j}{}' \int \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \frac{e^{-\beta U_N}}{Z} d^N\vec{r} =$$

$$= \left\langle \sum_{ij}{}' \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \right\rangle = \rho^{(2)}(\vec{r}', \vec{r})$$

$\rho^{(2)}(\vec{r}', \vec{r})$ IN THE HOMOGENEOUS CASE

LET'S CONSIDER

$$\left\langle \frac{1}{N\rho} \sum_{i,j}{}' \delta(\vec{r} - (\vec{r}_i - \vec{r}_j)) \right\rangle =$$

($\vec{r}_{ij}$; FUNCTION OF $\vec{r}$ AND WE AVERAGE OVER $\vec{r}_1 \dots \vec{r}_N$)

$$= \left\langle \frac{1}{N\rho} \sum_{i,j}{}' \delta(\vec{r} - \vec{r}_i + \vec{r}_j) \right\rangle =$$

$$= \left\langle \frac{1}{N\rho} \int \sum_{i,j}{}' \delta(\vec{r} - \vec{r}_i + \vec{r}') \cdot \delta(\vec{r}' - \vec{r}_j)\, d\vec{r}' \right\rangle =$$

($\rho^{(2)}(\vec{r} + \vec{r}', \vec{r}')$)

$$= \frac{1}{N\rho} \int \left\langle \sum_{i,j}{}' \delta((\vec{r} + \vec{r}') - \vec{r}_i) \cdot \delta(\vec{r}' - \vec{r}_j) \right\rangle d\vec{r}' =$$

$$= \frac{1}{N\rho} \int \rho^{(2)}(\vec{r}+\vec{r}', \vec{r}')\, d\vec{r}' =$$

BUT THE SYSTEM IS HOMOGENEOUS

THEREFORE $\rho^{(2)}(\vec{r}+\vec{r}', \vec{r}') = \rho^{(1)}(\vec{r})$

I.E IT IS INDEPENDENT OF $\vec{r}'$

$$= \frac{1}{N\rho} \int \rho^{(1)}(\vec{r})\, d\vec{r}' = \frac{V}{N\rho}\, \rho^{(1)}(\vec{r})$$

$$= \frac{1}{\rho^2}\, \rho^{(2)}(\vec{r}) = \frac{g^{(2)}(\vec{r})\cdot \cancel{\rho^2}}{\cancel{\rho^2}} =$$

$$= g^{(2)}(\vec{r})$$

WE PROVED THAT

$$g^{(2)}(\vec{r}) = \frac{1}{N\rho} \left\langle \sum_{ij}{}' \delta(\vec{r} - (\vec{r}_i - \vec{r}_j)) \right\rangle$$

$$= \frac{1}{N\rho} \left\langle \sum_{ij}{}' \delta(\vec{r} - \vec{r}_{ij}) \right\rangle \qquad \vec{r}'_{ij} = \vec{r}_i - \vec{r}_j$$

AVERAGE VALUE OF $g(\vec{r})$ OVER A VOLUME ΔV AROUND $\vec{r}$

$$\bar{g}(\vec{r}) = \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} g(\vec{r}')\, d\vec{r}'$$

$\bar{g}(\vec{r})$: DEPENDENT ON CHOSEN $\Delta V_{\vec{r}}$ (IF $\Delta V_{\vec{r}}$ IS A CUBIC BOX, $\bar{g}$ DEPENDS ON ITS POSITION $\vec{r}$)

$\Delta V_{\vec{r}}$: THIS IS TO REMIND THAT VOLUME ΔV IS AROUND $\vec{r}$

IN A MC SIMULATION:

$$\langle O(\vec{r}^N) \rangle = \frac{1}{L} \sum_{k=1}^{L} O(r_k^N) = \frac{1}{L} \sum_{k=1}^{L} O_k$$

L: OR EVERY A CERTAIN AMOUNT OF STEPS

$\sum_{k=1}^{L}$: SUM OVER MC STEPS

HENCE CONSIDER THE FOLLOWING OBSERVABLE

$$\bar{g}(\vec{r}; \vec{r}_1 \ldots \vec{r}_N) \equiv \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \left[\frac{1}{N\rho} \sum_{ij}{}' \delta(\vec{r}' - \vec{r}_{ij})\, d\vec{r}' \right]$$

$g(\vec{r})$ OF ONE CONFIGURATION (NO AVERAGE OVER $\frac{e^{-\beta U}}{Z}$)

ITS AVERAGE VALUE IS

$$\langle \bar{g} \rangle = \left\langle \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{iT}{}' \delta(\vec{r}' - \vec{r}_{iT})\, d\vec{r}' \right\rangle$$

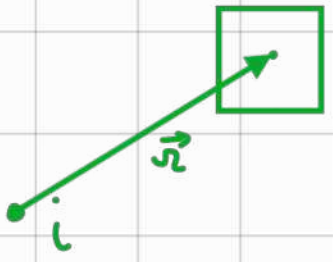
$$= \int \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{iT}{}' \delta(\vec{r}' - \vec{r}_{iT})\, d\vec{r}' \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_1 \ldots d\vec{r}_N d\vec{r}$$

$$= \frac{1}{L} \sum_{k=1}^{L} \underbrace{\frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{i,T}{}' \delta(\vec{r}' - \vec{r}_{iT}^{(k)})\, d\vec{r}'}_{\bar{g}_k} = \frac{1}{L} \sum_{k}^{L} \bar{g}_k$$

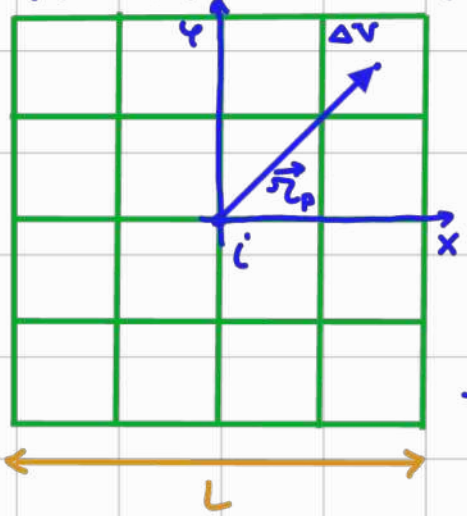
$\bar{g}_k$ IS OUR OBSERVABLE (AS $\mathcal{O}_k$) CALCULATED AT k-th MC STEP

$$\bar{g}_k(\vec{r}) = \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{i,T}{}' \delta(\vec{r}' - \vec{r}_{iT}^{(k)})\, d\vec{r}'$$

$$= \overbrace{\frac{1}{\Delta V \rho} \frac{1}{N} \sum_{i}}^{\text{AVERAGE OVER ALL PARTICLES}} \underbrace{\int_{\Delta V_{\vec{r}}} \sum_{T}{}' \delta(\vec{r}' - \vec{r}_{iT}^{(k)})\, d\vec{r}'}_{\#\text{ PARTICLES IN } \Delta V_{\vec{r}} \text{ A DISTANCE } \vec{r} \text{ FROM PARTICLE } i}$$



BOX CENTERED AT i OF VOLUME V DIVIDED INTO M SMALL BOXES OF VOLUME ΔV



$\vec{r}_p$ WITH $p = 1 \ldots M$ IS THE CENTER OF p-th BOX OF VOLUME ΔV

$\vec{r}_p = \Delta r\,(k\hat{x} + \ell\hat{y} + m\hat{z})$

k, ℓ, m INTEGERS

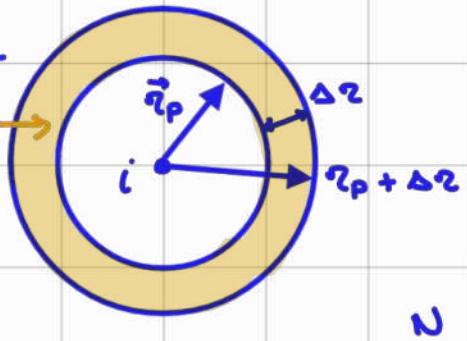
THUS OBTAINING

$\bar{g}_k(\vec{r}_p) \approx g_k(\vec{r}_p)$ IF ΔV SMALL ENOUGH

ISOTROPIC CASE

SPHERICAL SHELL OF THICKNESS Δr

$$\Delta V_p = 4\pi r_p^2 \Delta r$$

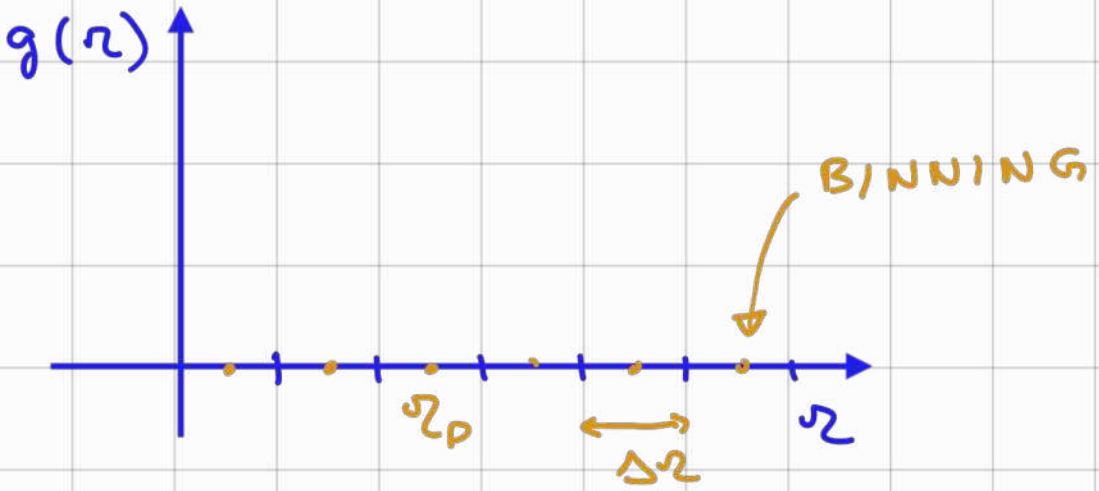


$$\overline{g}_k(r_p) = \underbrace{\frac{1}{4\pi r_p^2 \Delta r_p}}_{\text{AVERAGE NUMBER OF PARTICLES INSIDE } \Delta V_p} \frac{1}{N} \sum_i^N \int_{\Delta V_p} \overbrace{\sum_j{}' \delta(\vec{r} - \vec{r}_{ij})}^{\#\text{ PARTICLE INSIDE p-th SHELL}} d\vec{r} =$$

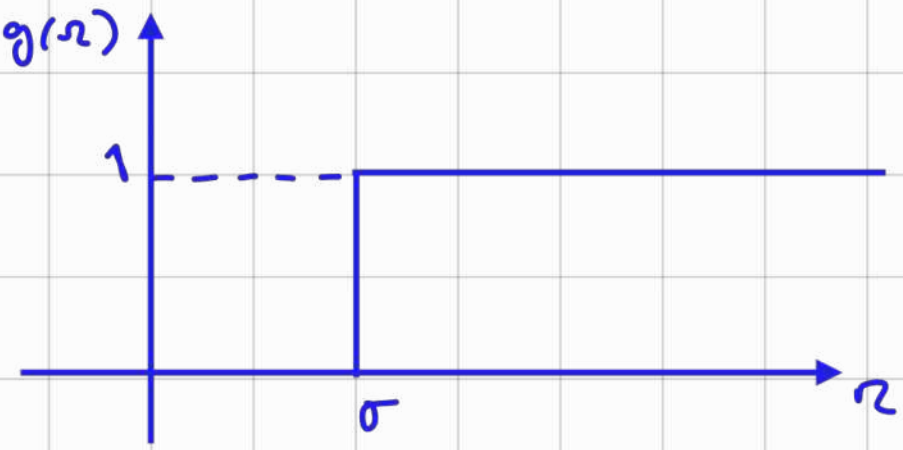
NOTE THAT THIS IS A SCALAR

$\overline{g}_k(r_p) \approx g_k(r_p)$ IS THE RADIAL DISTRIBUTION FUNCTION

IF Δr IS SMALL ENOUGH



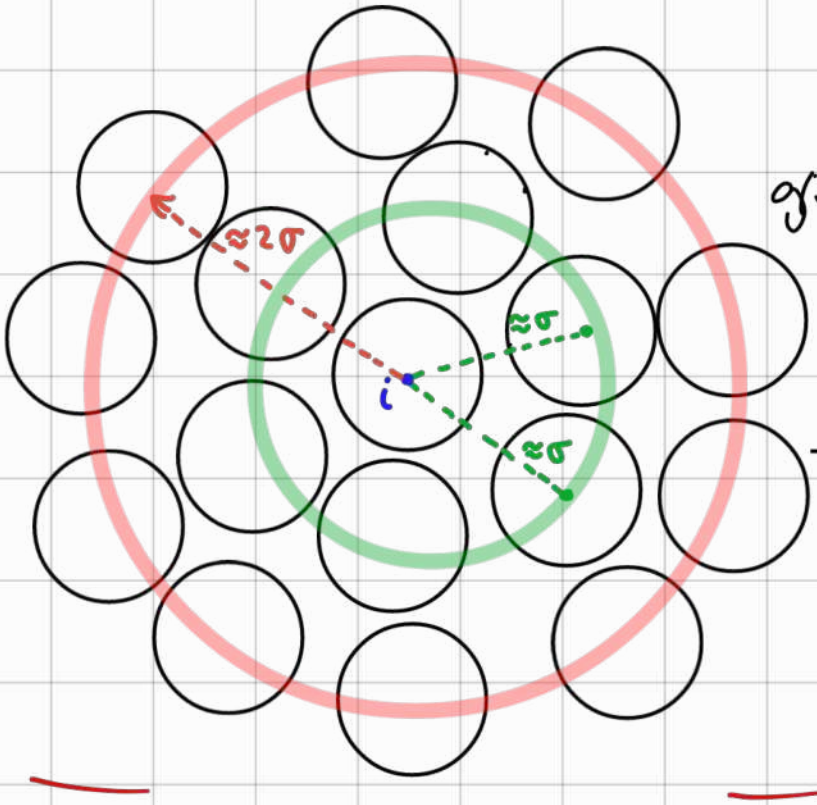
LOW CONCENTRATIONS

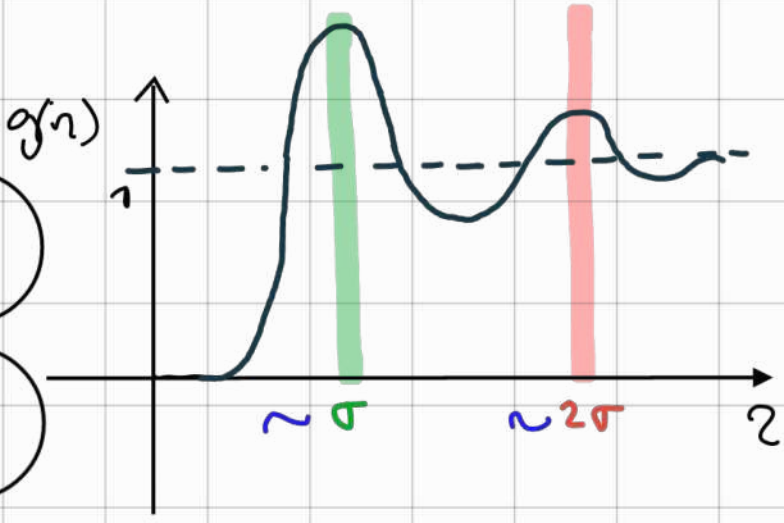


TODAY'S TOPICS 19/10/26

- CALCULATION OF ENERGY AND PRESSURE FROM $g(r)$
- TAIL CORRECTIONS
- STATIC STRUCTURE FACTOR AND MEAN SQUARED DISPLACEMENT

HIGH CONCENTRATION IN FLUID PHASE





CALCULATION OF ENERGY FROM g(r)

$$\langle U_N \rangle = U_{ex} = \int \overbrace{\frac{1}{2}\sum_{ij}' U_2(\vec{r}_{ij})}^{U_N} \frac{e^{-\beta U_N}}{Z} d^N\vec{r} =$$

$$= \frac{1}{2} N(N-1) \int U_2(\vec{r}_{12}) \frac{e^{-\beta U_N}}{Z} d\vec{r}_1 d\vec{r}_2 d^{(N-2)}\vec{r}$$

$$= \frac{1}{2} \int U_2(\vec{r}_{12}) \underbrace{\int N(N-1) \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r}}_{\rho^{(2)}(\vec{r}_1, \vec{r}_2)} d\vec{r}_1 d\vec{r}_2$$

$$= \frac{1}{2} \int U_2(\vec{r}_{12})\, \rho^{(2)}(\vec{r}_1, \vec{r}_2)\, d\vec{r}_1 d\vec{r}_2$$

BUT $\rho^{(2)}(\vec{r}_1, \vec{r}_2) = \rho^2\, g(\vec{r}_{12})$ ← HOMOGENEOUS CASE

$$= \frac{\rho^2}{2} \int U_2(\vec{r}_{12})\, g(\vec{r}_{12})\, d\vec{r}_1 d\vec{r}_2$$

IF WE CHANGE VARIABLES

$(\vec{r}_1, \vec{r}_2) \to (\vec{r}_2, \vec{r})$ WHERE

$\vec{r} = \vec{r}_{12} = \vec{r}_1 - \vec{r}_2$

$$= \frac{\rho^2}{2} \int v_2(\vec{r})\, g(\vec{r})\, d\vec{r}\, d\vec{r}_2 =$$

$$\langle U_N \rangle = \frac{V \rho^2}{2} \int v_2(\vec{r})\, g(\vec{r})\, d\vec{r} = U_{EX}$$

IF THE SYSTEM IS ISOTROPIC

$$= V \frac{\rho^2}{2} \int v_2(r)\, g(r)\, 4\pi r^2\, dr$$

↑ RADIAL DISTRIBUTION FUNCTION

TAIL CORRECTIONS

IN A SIMULATION $v_{ij}(r) \to v_{ij}^c(r)$

WHERE $v_{ij}^c = \begin{cases} v_{ij}(r) & r \le r_c \\ 0 & r > r_c \end{cases}$

HENCE

$$U_{EX} = \langle U_N \rangle = \frac{1}{2} V \rho^2 \int_0^\infty g(r)\, v_2(r)\, 4\pi r^2\, dr$$

U_{EX}^c THIS IS WHAT WE HAVE FROM SIMULATIONS, I.E. $\frac{1}{2}\sum_{i,j}' v_2^c(r_{ij})$

$$= \underbrace{\frac{1}{2} V \rho^2 \int_0^{r_c} g(r)\, v_2(r)\, 4\pi r^2\, dr}_{} +$$

$$+ \underbrace{\frac{1}{2} V \rho^2 \int_{r_c}^\infty g(r)\, v_2(r)\, 4\pi r^2\, dr}_{\text{TAIL CORRECTION}}$$

Q. CAN WE ESTIMATE SOMEHOW THE TAIL CORRECTION?

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} g(r)\, u_2(r)\, 4\pi r^2\, dr$$

↑ TAIL CORRECTION

IF r_c IS LARGE ENOUGH FOR $r \geq r_c$ $g(r) \approx 1$

HENCE

THIS INTEGRAL CAN BE CALCULATED

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} u_2(r)\, 4\pi r^2\, dr$$

EXAMPLE LENNARD-JONES SYSTEM

$$u_2(r) = u_{LJ}(r) = A\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

SO THAT

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} 4\pi r^2 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] =$$

$$= V \frac{8\pi}{3} \varepsilon \rho^2 \sigma^3 \left[\frac{1}{3}\left(\frac{\sigma}{r_c}\right)^9 - \left(\frac{\sigma}{r_c}\right)^3\right]$$

WE CAN DO ALSO BETTER

ENERGY FROM SIMULATION

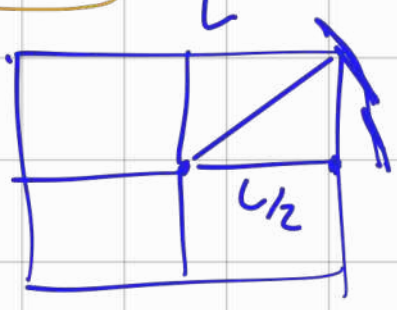
$$U_{ex} = \frac{1}{2} V \rho^2 \int_0^{r_c} g(r)\, u_2(r)\, 4\pi r^2\, dr +$$

TAIL CORRECTION

$$\frac{1}{2} V \rho^2 \int_{r_c}^{L/2} g(r)\, u_2(r)\, 4\pi r^2\, dr + \approx 1$$

$$\frac{1}{2} V \rho^2 \int_{L/2}^{+\infty} g(r)\, u_2(r)\, 4\pi r^2\, dr =$$

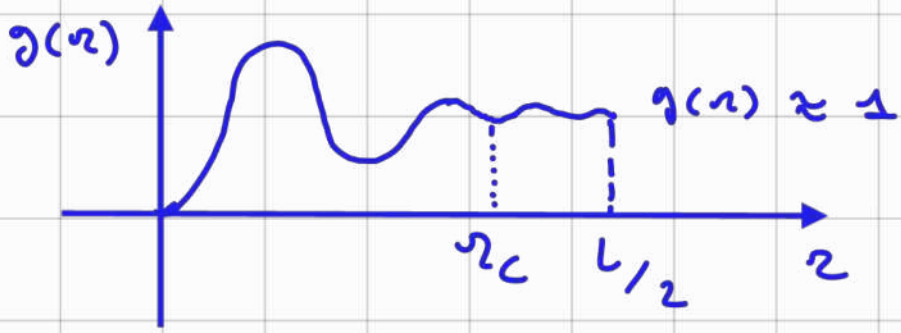
TAIL CORRECTION



$$= U_{ex}^{C} + U_{ex}^{TAIL}$$

WHERE $g(r)$ CALCULATED FROM SIMULATIONS

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{L/2} g(r)\, u_L(r)\, 4\pi r^2\, dr + \frac{1}{2} V \rho^2 \int_{L/2}^{+\infty} u_L(r)\, 4\pi r^2\, dr$$



CALCULATION OF PRESSURE FROM $g(r)$

$\mathcal{H} = K + U$

$$P = -\frac{\partial F}{\partial V} = k_B T \frac{\partial \ln Z}{\partial V} \qquad F = -k_B T \ln Q \quad \text{BUT}$$

$$Z = Z_{ID} \cdot Z_{ex} \qquad Z_{ID} = V^N \qquad Q = \frac{\Lambda^{-3N}}{N!} Z$$

$$Z_{ex} = \int e^{-\beta U_N(\vec{r}_1 \ldots \vec{r}_N)} d^N r \qquad \text{WHERE} \quad \Lambda = \sqrt{\frac{2\pi\hbar^2}{m k_B T}}$$

IT DOES NOT DEPEND ON V

SCALED COORDINATES:

$$\vec{r}_i = V^{1/3} \vec{s}_i$$

$$P = k_B T \frac{\partial}{\partial V} (\ln Z_{ID} + \ln Z_{ex}) = k_B T \frac{\partial \ln V^N}{\partial V} + k_B T \frac{\partial \ln Z_{ex}}{\partial V} = \underbrace{k_B T \rho}_{P_{ID}} + \underbrace{k_B T \frac{\partial \ln Z_{ex}}{\partial V}}_{P_{ex}}$$

$$P = P_{ID} + P_{ex}$$

$U_N = U_N(V^{1/3} s_1 \ldots V^{1/3} s_N)$

$$P_{ex} = k_B T \frac{1}{Z_{ex}} \frac{\partial Z_{ex}}{\partial V} = \frac{k_B T}{Z_{ex}} \frac{\partial}{\partial V} \left(\int_0^1 e^{-\beta U_N} d^N s \right) =$$

$$= -\cancel{k_B T}\,\beta \int \frac{\partial U_N}{\partial V}\,\frac{e^{-\beta U_N} V^N}{Z_{ex}\, V^N}\, d\vec{s}^N = -\int \frac{\partial U_N}{\partial V}\,\frac{e^{-\beta U_N}}{Z}\, d\vec{s}^N =$$

(*) NOTE THAT WE CAN BRING $\frac{\partial}{\partial V}$ INSIDE THE INTEGRAL SINCE THE INTEGRATION DOMAIN DOES NOT DEPEND ON V IF WE USE SCALE COORDS.

$$= -\left\langle \frac{\partial U_N}{\partial V} \right\rangle$$

$$P = P_{ID} - \underbrace{\left\langle \frac{\partial U_N}{\partial V} \right\rangle}_{P_{EX}}$$

$$U_N = \frac{1}{2}\sum_{i,j}{}' u_2(\vec{r}_{ij}) \qquad V = L^3 \Rightarrow \frac{\partial}{\partial V} = \frac{1}{3L^2}\frac{\partial}{\partial L}$$

$$P_{EX} = -\left\langle \frac{1}{3L^2}\frac{\partial U_N}{\partial L} \right\rangle = -\left\langle \frac{1}{3L^2}\frac{\partial}{\partial L}\left(\frac{1}{2}\sum_{ij}{}' u_2(\vec{r}_{ij})\right)\right\rangle =$$

$$= -\left\langle \frac{1}{6L^2}\sum_{ij}{}' \underbrace{\frac{\partial u_2}{\partial \vec{r}_{ij}}}_{-\vec{f}_{ij}} \cdot \frac{\partial \vec{r}_{ij}}{\partial L} \right\rangle = \left\langle \frac{1}{6L^3}\sum_{ij}{}' \vec{f}_{ij}\cdot\vec{r}_{ij} \right\rangle =$$

($\vec{s}_{ij} = \frac{\vec{r}_{ij}}{L}$)

NOTE THAT FOR A SYSTEM WITH PBC WE CAN RESTRICT THE CALCULATION TO "FIRST" BOX, SO THAT:

$$U_N = \frac{1}{2}\sum_{\vec{n}}\sum_{i,j} u_2(\vec{r}_i - \vec{r}_j + L\vec{n})$$

WHERE $\vec{n} = (n_x, n_y, n_z)$

WITH n_α INTEGERS

HENCE

$$\frac{\partial U_N}{\partial L} = \frac{1}{2}\sum_{ij}{}' \frac{\partial u_2(\vec{r}_i - (\vec{r}_j + L\vec{n}))}{\partial L} =$$

$$= \frac{1}{2}\sum_{ij}{}' \frac{\partial u_2}{\partial \vec{r}_{ij}} \cdot \frac{\partial(\vec{r}_i - (\vec{r}_j + L\vec{n}))}{\partial L}$$

BUT

$$\frac{\partial(\vec{r}_i - (\vec{r}_j + L\hat{n}))}{\partial L} = \frac{\vec{r}_i - (\vec{r}_j + L\vec{n})}{L} = \frac{\vec{r}_{ij}}{L}$$

WHERE j RUNS OVER ALSO PERIODIC PARTICLES

(FORCE ACTING BETWEEN I AND J)

$$= \frac{1}{6V} \left\langle \sum_{ij}{}' \vec{f}_{ij} \cdot \vec{r}_{ij} \right\rangle$$

SIMILAR EXPRESSION FOR MOLECULAR PRESSURE

$$P = \rho k_B T + \overbrace{\frac{1}{6V} \left\langle \sum_{ij}{}' \vec{f}_{ij} \cdot \vec{r}_{ij} \right\rangle}^{P_{EX}}$$

WE CAN NOW WRITE P_{EX} IN TERMS OF $g(\vec{r})$

ALL TERMS OF $\sum_{ij}{}'$ ARE EQUAL

$$P_{EX} = \frac{N(N-1)}{6V} \int \underbrace{\frac{e^{-\beta U_N}}{Z}} \vec{f}_{12} \cdot \vec{r}_{12} \, d\vec{r}_1 d\vec{r}_2 \underline{d\vec{r}^{(N-2)}} =$$

$$= \frac{1}{6V} \int \underline{\rho^{(2)}(\vec{r}_1, \vec{r}_2)} \vec{f}_{12} \cdot \vec{r}_{12} \, d\vec{r}_1 d\vec{r}_2$$

CHANGE OF VARIABLES

$(\vec{r}_1, \vec{r}_2) \rightarrow (\vec{r}, \vec{r}_2)$ WHERE $\vec{r} = \vec{r}_1 - \vec{r}_2$

$$= \frac{1}{6V} \int \rho^{(2)}(\vec{r} + \vec{r}_2, \vec{r}_2) \vec{f}(\vec{r}) \cdot \vec{r} \, d\vec{r} \, \underline{d\vec{r}_2} =$$

IF SYSTEM IS HOMOGENEOUS

$\rho^{(1)}(\vec{r} + \vec{r}_2, \vec{r}_2) = \rho^{(2)}(\vec{r})$ HENCE

$$= \frac{1}{6} \int \rho^{(2)}(\vec{r}) \, \vec{f}(\vec{r}) \cdot \vec{r} \, d\vec{r}$$

BUT $\rho^{(2)}(\vec{r}) = \rho^2 g(\vec{r})$

$$= \frac{\rho^2}{6} \int \vec{f}(\vec{r}) \cdot \vec{r} \, g(\vec{r}) \, d\vec{r} \leftarrow$$

IF THE SYSTEM IS ISOTROPIC

$$P_{EX} = \frac{\rho^2}{6} \int_0 f(r) \, g(r) \, 4\pi r^3 \, dr \leftarrow$$

TAIL CORRECTIONS P_{SIM} FROM SIMULATIONS

$$P_{EX} = \underbrace{\frac{\rho^2}{6}\int_0^{r_c} f(r)\, g(r)\, 4\pi r^3 dr}_{P_{SIM}} + \underbrace{\frac{\rho^2}{6}\int_{r_c}^{+\infty} f(r)\, \cancel{g(r)}^{\approx 1}\, 4\pi r^3 dr}_{\text{FOR LJ WE CAN CALCULATE THIS ANALITICALLY}} = P_{TAIL}$$

(•) NOTE THAT

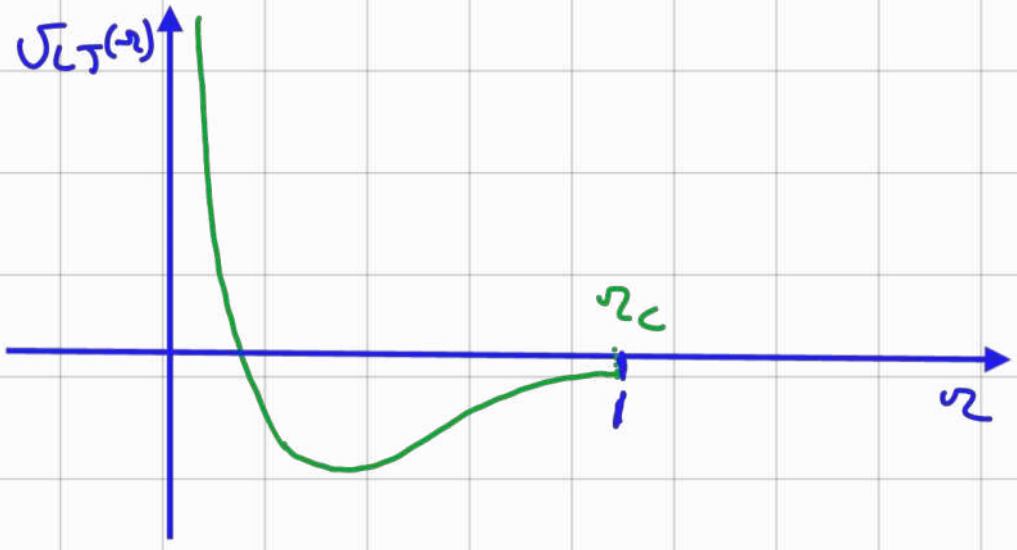
WE CAN CALCULATE THIS FROM SIMULATIONS

$$P_{SIM} = \frac{\rho^2}{6}\int_0^{r_c} f(r)\, g(r)\, 4\pi r^3 dr = \frac{1}{6V}\left\langle \sum_{ij}{}' \vec{F}_{ij}^{c} \cdot \vec{r}_{ij} \right\rangle$$

THE ABOVE TAIL CORRECTION APPLIES IF WE WANT TO COMPARE OUR RESULTS WITH A SYSTEM WITHOUT CUTOFF, I.E. WE HAVE: $P = P_{SIM} + P_{TAIL}$

Q. BUT WHAT IS THE PRESSURE OF A SYSTEM WITH TRUNCATED INTERACTIONS?

FOR EXAMPLE CONSIDER A TRUNCATED LJ SYSTEM



AROUND r_c ONE HAS

$$v_{LT}(r) \approx \theta(r_c - r)\, v_{LT}(r_c)$$

WHERE

$$\theta(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$f(r) = -\frac{dv_2}{dr} = v_{LT}(r_c)\,\delta(r - r_c)$$

SINCE

$$\frac{d}{dx}\theta(x) = \delta(x)$$

HENCE WE HAVE

$$P_{IMP} = \frac{1}{6}\rho^2 \int_{r_c-\varepsilon}^{\infty} v_{LT}(r_c)\,\delta(r - r_c)\,4\pi r^3\,dr =$$

$$= \frac{1}{6}\rho^2\, v_{LT}(r_c)\,4\pi r_c^3$$

"IMPULSIVE" CORRECTION TO PRESSURE

TODAY'S TOPICS 23/10/26

- STATIC STRUCTURE FACTOR
- SETTING-UP A SIMULATION: INITIAL CONFIGURATION
- TRICKS OF THE TRADE: LINKED CELL LISTS AND NEAREST NEIGHBOUR LISTS
- ISOBARIC ENSEMBLE

STATIC STRUCTURE FACTOR

1 POINT DENSITY

$$\rho(\vec{r}) = \left\langle \sum_i \delta(\vec{r}-\vec{r}_i) \right\rangle$$

$$\vec{k} = \frac{2\pi}{L}(n_1, n_2, n_3)$$

THEN

$$\langle \rho(\vec{k}) \rangle = \overbrace{\int \rho(\vec{r}) e^{-i\vec{k}\cdot\vec{r}}\, d\vec{r}}^{\text{FOURIER TRANSFORM}} = \overbrace{\rho(\vec{k})}^{} = \left\langle \sum_i e^{-i\vec{k}\cdot\vec{r}_i} \right\rangle \quad \text{(ENSEMBLE AVERAGE)}$$

WHERE

$$\rho(\vec{k}) = \sum_i^N e^{-i\vec{k}\cdot\vec{r}_i}$$

WE DEFINE

$$S(\vec{k}) \equiv \frac{1}{N} \langle \rho(\vec{k})\, \rho^*(\vec{k}) \rangle$$

ENSEMBLE AVERAGE

$S(\vec{k})$ CAN BE MEASURED EXPERIMENTALLY THROUGH SCATTERING EXPERIMENTS

BUT

$$S(\vec{k}) = \frac{1}{N}\left\langle \sum_{i,j} e^{-ik(\vec{r}_i-\vec{r}_j)} \right\rangle = 1 + \frac{1}{N}\left\langle \sum_{i,j}{}' e^{-i\vec{k}\cdot(\vec{r}_i-\vec{r}_j)} \right\rangle =$$

($i \neq j$)

$$= 1 + \frac{1}{N}\left\langle \int \sum_{i,j}{}' e^{-ik\cdot(\vec{r}-\vec{r}')} \delta(\vec{r}-\vec{r}_i)\, \delta(\vec{r}'-\vec{r}_j)\, d\vec{r}\, d\vec{r}' \right\rangle =$$

$$= 1 + \frac{1}{N}\int d\vec{r}\, d\vec{r}'\, e^{-i\vec{k}\cdot(\vec{r}-\vec{r}')} \overbrace{\left\langle \sum_{i,j}{}' \delta(\vec{r}-\vec{r}_i)\, \delta(\vec{r}-\vec{r}_j) \right\rangle}^{\rho^{(2)}(\vec{r},\vec{r}')} =$$

$$= 1 + \frac{1}{N}\int d\vec{r}\, d\vec{r}'\, e^{-ik(\vec{r}-\vec{r}')}\, \rho^{(2)}(\vec{r},\vec{r}') =$$

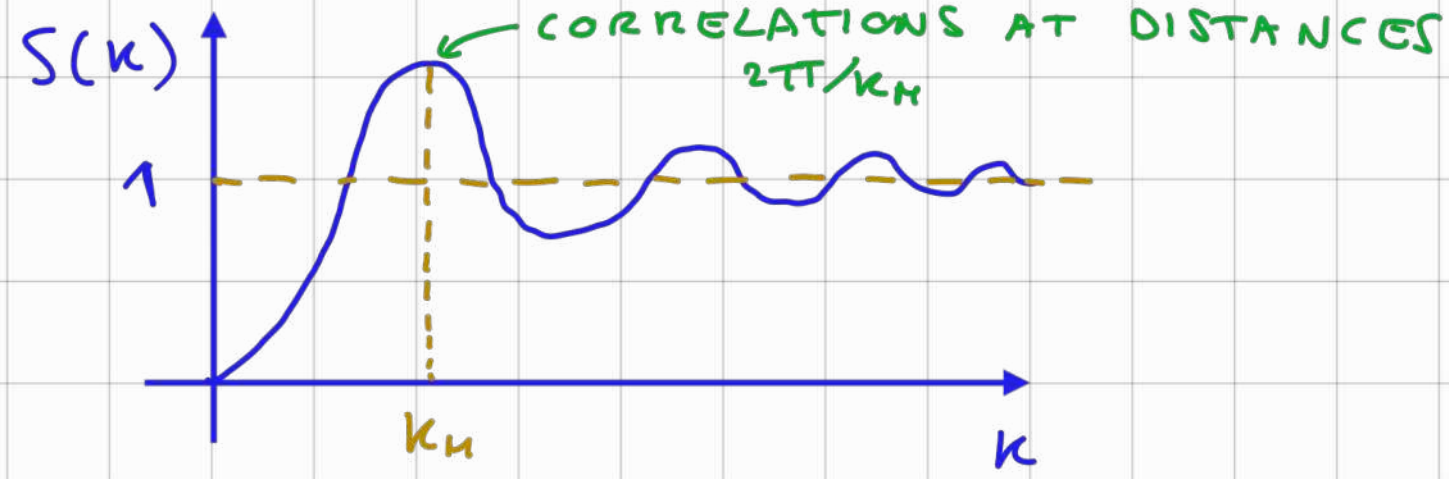
$\rho^{(2)} = \rho^2 g^{(2)}(\vec{r},\vec{r}')$

ASSUMING THE SYSTEM HOMOGENEOUS

$$= 1 + \frac{\rho^2}{N} V \int d\vec{r}\, e^{-i\vec{k}\cdot\vec{r}}\, g(\vec{r}) =$$

$$= 1 + \rho \underbrace{\int d\vec{r}\, e^{-i\vec{k}\cdot\vec{r}}\, g(\vec{r})}_{\text{FOURIER TRANSFORM OF } g(\vec{r})}$$

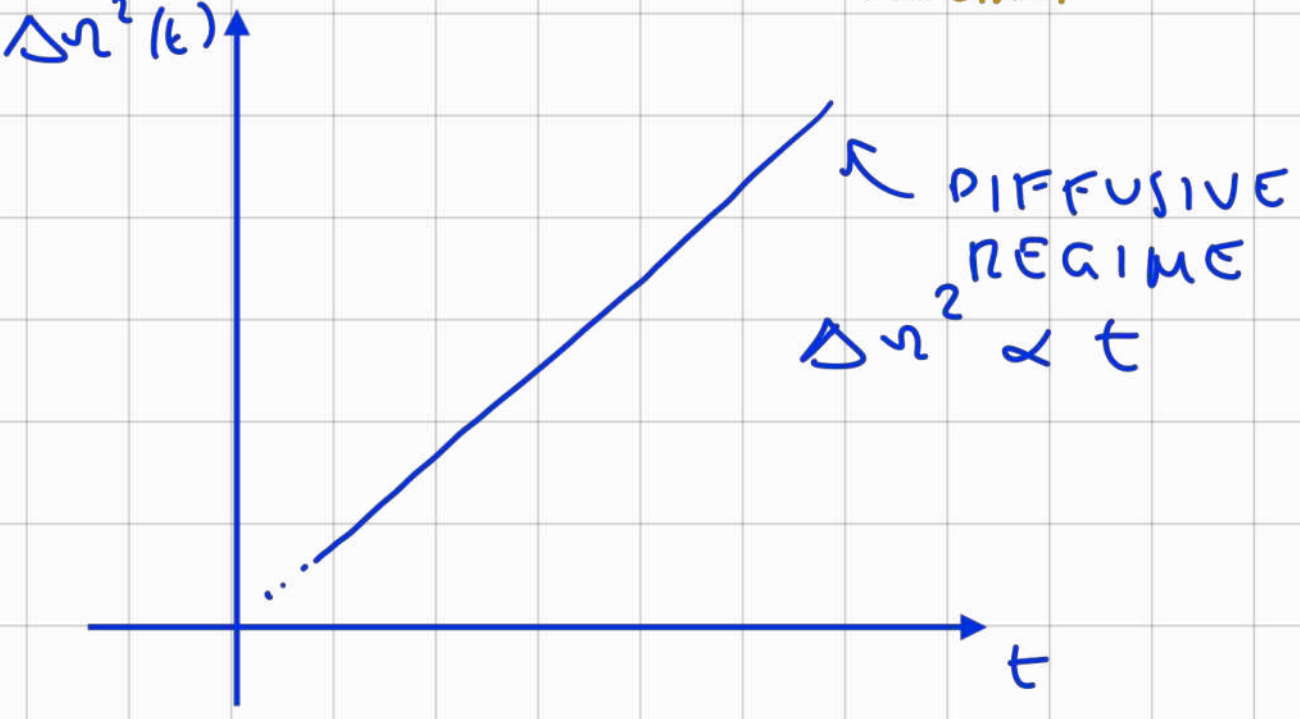
IN THE ISOTROPIC CASE



MEAN SQUARED DISPLACEMENT

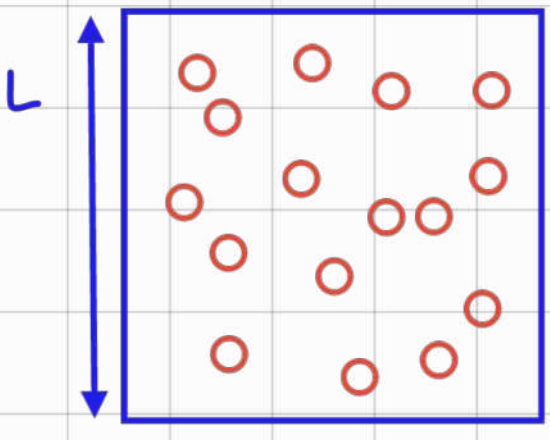
$$\Delta r^2(t) \equiv \frac{1}{N} \sum_{i=1}^{N} \| \vec{r}_i(t) - \vec{r}_i(0) \|^2$$

- $\Delta r^2(t)$: FUNCTION OF t
- $\vec{r}_i(t)$: PASSO MC
- $\vec{r}_i(0)$: POSIZIONI INIZIALI



DIFFUSIVE REGIME

$\Delta r^2 \propto t$



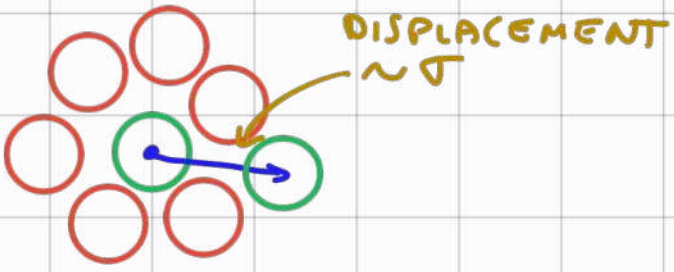
$\sqrt{\Delta r^2} \gtrsim L$

THE SYSTEM CAN BE CONSIDERED EQUILIBRATED SINCE ALL PARTICLES HAVE MOVED ENOUGH I.E. HAVE FULLY EXPLORED THE CONFIGURATION SPACE

WE COULD ALSO USE A LESS STRINGENT CONDITION, I.E.

TYPICAL PARTICLE SIZE

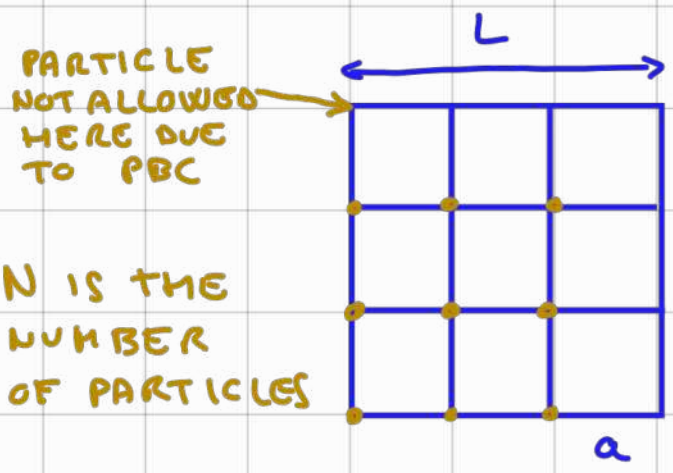
$$\sqrt{\Delta r^2} \gtrsim \sigma$$



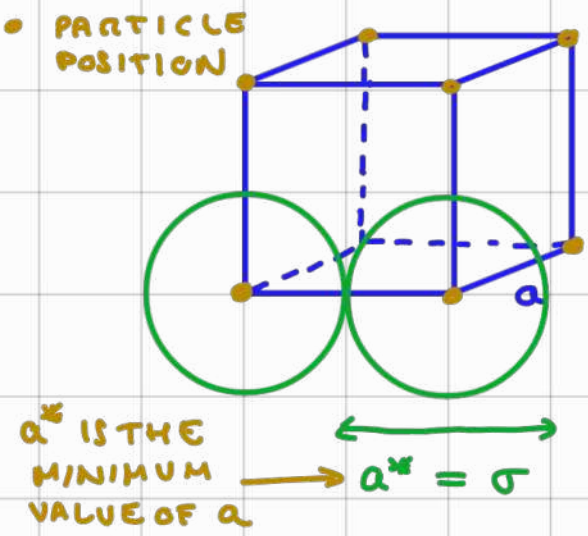
MAIN CONTRIBUTION TO SPATIAL CORRELATIONS COMES FROM FIRST PEAK OF $S(R)$ WHICH ROUGHLY CORRESPONDS TO A DISTANCE EQUAL TO σ

INITIAL CONFIGURATION

CUBIC LATTICE



EACH PARTICLE HAS DIAMETER σ



$$\phi = \frac{N v_0 n}{L^3} = \frac{n \left(\frac{L}{a}\right)^3 v_0}{L^3}$$

(N = $n\left(\frac{L}{a}\right)^3$)

WHERE v_0 IS THE VOLUME OF A PARTICLE AND n IS THE NUMBER OF PARTICLES PER UNIT CELL ($n=1$ IN THIS CASE)

$$= n \frac{v_0}{a^3} = \frac{n v_0}{a^3}$$

BUT $v_0 = \frac{\pi}{6}\sigma^3$ IF PARTICLE IS SPHERICAL

HENCE CONSIDERING THAT $a^* = \sigma$

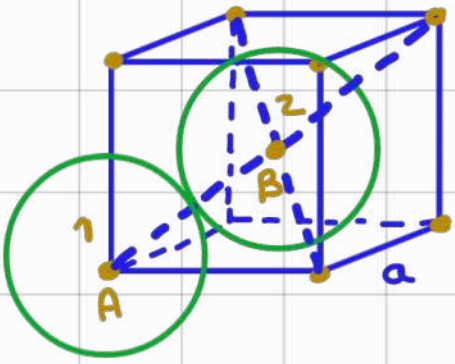
$$\phi^* = \{\text{MAXIMUM } \phi\} = \frac{v_0}{(a^*)^3} = \frac{\pi}{6}\frac{\sigma^3}{\sigma^3} = \frac{\pi}{6}$$

IF WE WANT A LOWER $\phi = \phi'$ IT IS JUST A MATTER TO EXPAND THE BOX AND SCALE PARTICLE POSITIONS BY A FACTOR ξ SUCH THAT

$$L' = \xi L \qquad \vec{r}_i' = \xi \vec{r}_i$$

WITH $\xi = \left(\frac{\phi^*}{\phi'}\right)^{1/3}$ I.E. $\phi' = \frac{\phi^*}{\xi^3}$

HIGHER ϕ: BODY CENTERED CUBIC (BCC)



$$\phi^* = \frac{n\, v_0}{(a^*)^3}$$

WHERE IN THIS CASE

$n = 2$ AND

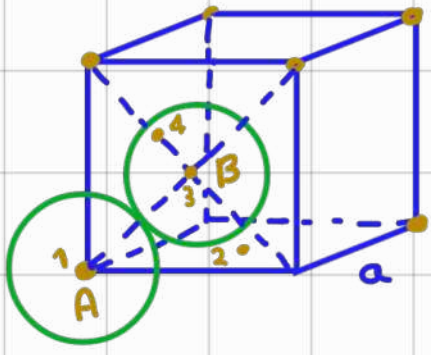
a^* IS SUCH THAT

$$\overline{AB} = \sigma = \frac{\sqrt{3}\, a^*}{2} \quad \text{I.E. } a^* = \frac{2}{\sqrt{3}}\sigma$$

SO THAT

$$\phi^* = \frac{2\,\frac{\pi}{6}\,\sigma^3}{\left(\frac{2}{\sqrt{3}}\right)^3 \sigma^3} = \frac{\pi}{3}\left(\frac{\sqrt{3}}{2}\right)^3 \approx 0.68$$

EVEN HIGHER ϕ : FACE CENTERED CUBIC (FCC)



$$\phi^* = \frac{n\, v_0}{(a^*)^3}$$

WHERE IN THIS CASE

$n = 4$ AND

a^* IS SUCH THAT

$$\overline{AB} = \sigma = \frac{\sqrt{2}}{2}\, a^* \quad \text{I.E. } a^* = \frac{2}{\sqrt{2}}\sigma$$

SO THAT

$$\phi^* = \frac{4\,\frac{\pi}{6}\,\sigma^3}{\left(\frac{2}{\sqrt{2}}\right)^3 \sigma^3} = 2\left(\frac{\sqrt{2}}{2}\right)^3 \frac{\pi}{3} \approx 0.74$$

LINKED CELL LISTS

IN A METROPOLIS MC A MC STEP CONSISTS OF ATTEMPTING TO MOVE N PARTICLE

FOR EACH PARTICLE

$$acc(o \to n) = \min\{1, e^{-\beta \Delta U}\}$$

$\Delta U = U_i' - U_i$ WHERE

$U_i = \sum_j u_2(\vec{r}_{ij})$ WHICH REQUIRES N CALCULATIONS OF u_2 (NOTE THAT EVEN IF ONLY WE NEED TO CALCULATE $|\vec{r}|$)

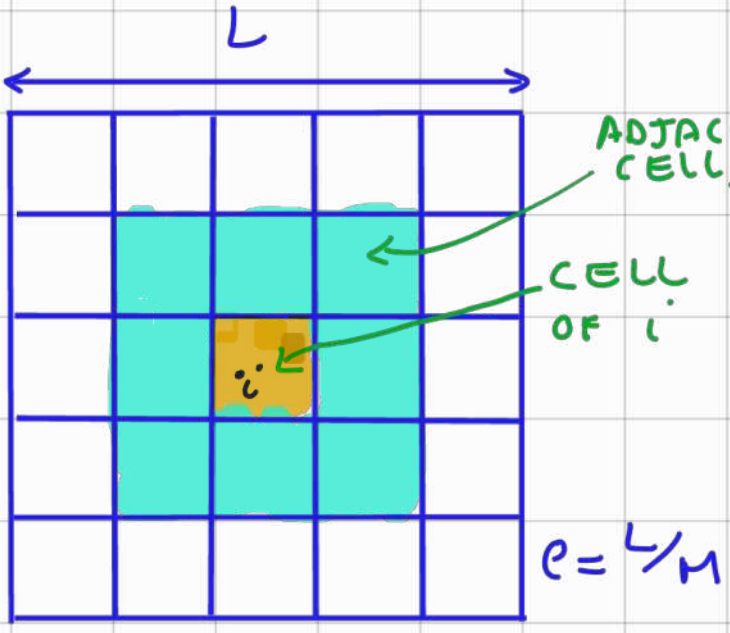
HENCE FOR A MC STEP COMPLEXITY IS $O(N \times N) = O(N^2)$

WE WANT TO REDUCE THE NUMBER OF TIMES WE HAVE TO CALCULATE u_2

LET'S CONSIDER M^3 CUBIC CELLS SUCH THAT $L/M \geq r_c$ ← CUTOFF RADIUS

I.E. M IS THE LARGEST INTEGER SUCH THAT

$$\ell = \frac{L}{M} \geq r_c$$



TO CALCULATE U_i AND U_i' WE CAN JUST CONSIDER PARTICLES IN THE 27 CELLS (I.E. THE CELL TO WHICH PARTICLE i BELONGS AND THE 26 ADJACENT CELLS)

WITH THIS STRATEGY FOR EACH MC MOVE WE HAVE TO CALCULATE $\tilde{N}$ TIMES U_L WITH

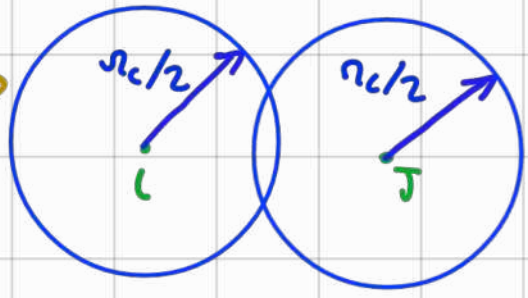
$$\tilde{N} \approx N\, 27 \cdot r_c^3 \rho = N \underbrace{27 r_c^3 \rho}_{\text{THIS IS A CONSTANT IF WE INCREASE BOX SIZE KEEPING FIXED } \rho} = O(N)$$

AFTER PARTICLE i MOVE, IF CELL TO WHIC IT BELONGS CHANGES, CELL LISTS MUST BE UPDATED

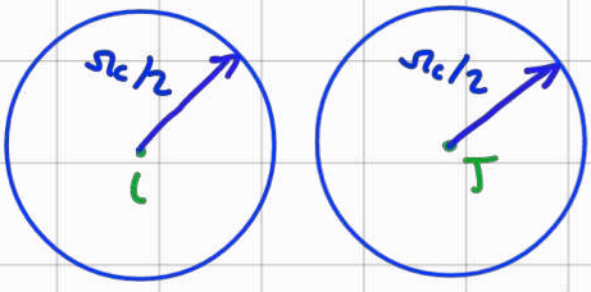
NEAREST NEIGHBOUR LISTS (NNL)

CONSIDER TWO PARTICLES i AND j AND OBSERVE THAT:

SPHERE OF DIAMETER r_c (CUTOFF RADIUS)



THE TWO SPHERES OVERLAP HENCE I AND J INTERACT

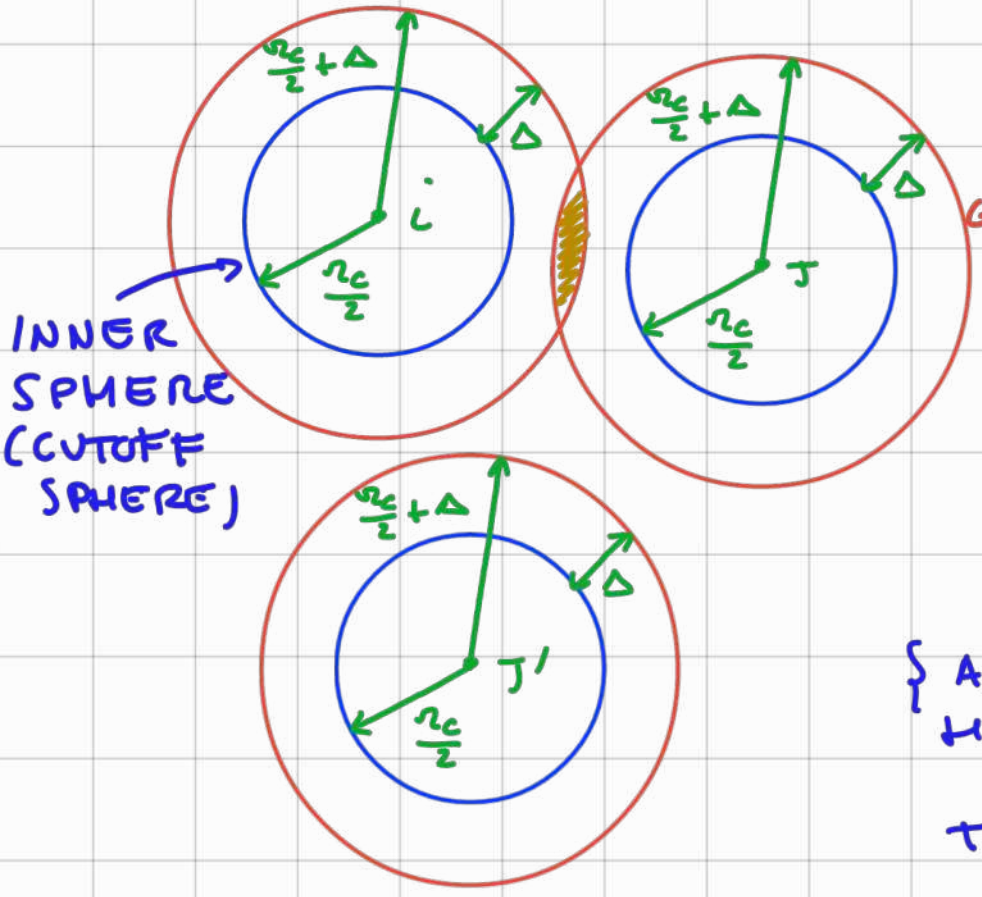


THE TWO SPHERES DO NOT OVERLAP HENCE i AND j DO NOT INTERACT

AT A CERTAIN TIME t_0 (MC STEP) WE BUILD THE NNL OF ALL PARTICLES (STARTING FROM FIRST SIMULATION STEP)

CONSIDER THE BUILDING OF NNL OF A PARTICLE i

j AND j' ARE TWO PARTICLES OTHER THAN i



INNER SPHERE (CUTOFF SPHERE)

OUTER SPHERE

$NNL(i) =$

{ LIST OF ALL PARTICLES BELONGING TO NNL OF i } =

{ ALL PARTICLES WHICH HAVE OUTER SPHERES OVERLAPPING WITH THAT OF i }

HENCE IN THIS CASE:

$j \in NNL(i)$

SINCE SPHERES OF RADIUS $\frac{r_c}{2}+\Delta$ OVERLAP

$j' \notin NNL(i)$

SINCE SPHERES OF RADIUS $\frac{r_c}{2}+\Delta$ DO NOT OVERLAP

FOR $t > t_0$ INNER SPHERE ARE PLACED WITH THEIR CENTER AT PARTICLE POSITION WHILE OUTER SPHERES KEEP FIXED

AS LONG AS EACH INNER SPHERE REMAIN INSIDE ITS CORRESPONDING OUTER SPHERE WE CAN CALCULATE U_i AND U_i' BY CONSIDERING $\forall i$ THE PARTICLES $\in NNL(i)$

SINCE IF TWO INNER SPHERES ARE INSIDE TWO NON OVERLAPPING OUTER SPHERES THEY CAN NOT OVERLAP (AND CORRESPONDING PARTICLES CAN NOT INTERACT)

AN INNER SPHERE ESCAPE IF $\| \vec{r}_i(t) - \vec{r}_i(t_0) \| \geq \Delta$

IF THIS HAPPENS ALL NNL MUST BE REBUILT

Δ IS A PARAMETER WHICH MUST BE OPTIMIZED



AND IF η_{LL} IS THE EFFICIENCY OF LL

$$\frac{n_{NNL}}{n_{LL}} = \frac{\{\#\text{ PARTICLES IN THE 27 LL CELLS}\}}{\{\#\text{ PARTICLES IN THE NNL(i)}\}}$$

↑ SINCE FOR EACH PARTICLE WE NEED TO TEST IF $r < r_c$

$$\frac{\rho\, r_c^3\, 27}{\frac{4\pi}{3} r_c^3\, \rho} \approx \frac{27 \cdot 3}{4\pi} \approx 6$$

↑ IF $\Delta \ll r_c$

NPT ENSEMBLE

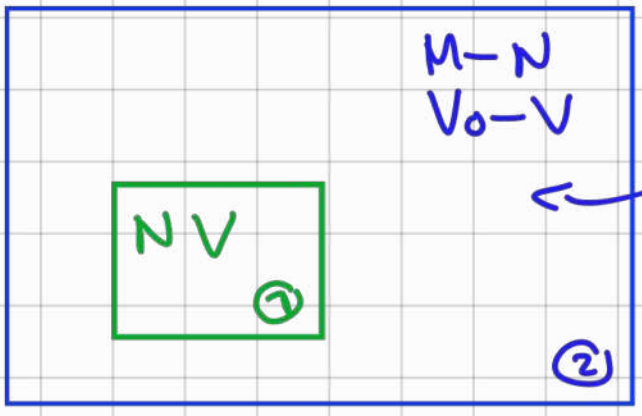
IF ρ IS A GENERIC DISTRIBUTION

$$\rho(o) \cdot \pi(o \to n) = \rho(n)\, \pi(n \to o)$$

WHERE $\pi(o \to n) = \underbrace{\alpha(o \to n)}_{\text{TRIAL MOVE}}\, acc(o \to n)$

WITH $$acc(o \to n) = \begin{cases} \rho(n)/\rho(o) & \text{IF } \rho(n) \leq \rho(o) \\ 1 & \text{OTHERWISE} \end{cases}$$

GOAL: FIND ρ FOR NPT ENSEMBLE



WEAKLY INTERACTING SYSTEMS

$$E_{TOT} = E_1 + E_2$$

IF THE VOLUMES OF ① AND ② ARE FIXED

$$Q = \int \frac{d^{M-N}q}{\Lambda^{3(M-N)}(M-N)!}\ \text{(②)} \int \frac{dq^N\, e^{-\beta U_N}}{\Lambda^{3N} N!}\ \text{(①)} \qquad U_N = \frac{1}{2}\sum_{ij}{}' U_2(\vec{r}_{ij})$$

WHERE $\Lambda = \sqrt{\frac{2\pi\hbar^2}{m k_B T}}$ ← THERMAL WAVELENGTH

SCALED COORDINATES

$V' = V_0 - V$

$\vec{q}_i = V^{1/3}\vec{\xi}_i$ FOR $i \in$ ①

$\vec{q}_i = (V')^{1/3}\vec{\xi}_i$ FOR $i \in$ ②

$$Q = \frac{(V')^{M-N} V^N}{\Lambda^{3M}(M-N)!N!} \underbrace{\int d^{M-N}s}_{=1} \int d^{3N}s \, e^{-\beta U_N} =$$

$$= \frac{(V')^{M-N} V^N}{\Lambda^{3M}(M-N)!N!} \int d^{N}s \, e^{-\beta U_N}$$

IF THE VOLUME IS ALLOWED TO CHANGE

$$Q = \int dV \frac{(V')^{M-N} V^N}{\Lambda^{3M}(M-N)!N!} \int d^{N}s \, e^{-\beta U_N}$$

↑ WE HAVE ALSO TO SUM OVER ALL POSSIBLE VOLUMES

FOR NVT ENSEMBLE

$$Z = \int e^{-\beta U_N} d^N q \qquad \rho \propto e^{-\beta U_N}$$

HENCE WE HAVE

$$\rho(\underbrace{V, \vec{s}_1, \ldots \vec{s}_N}_{\text{STATE OF SYSTEM}}) \propto e^{-\beta U_N} V^N (V_0 - V)^{M-N} \qquad V' = V_0$$

NOTE THAT TO IMPLEMENT A METROPOLIS MC WE NEED ρ UP TO A CONSTANT SINCE ACCEPTANCE PROBABLITY DEPENDS ON $\rho(n)/\rho(o)$

CONSIDER THE TERM

$$(V_0 - V)^{M-N} = V_0^{M-N} \left(1 - \frac{V}{V_0}\right)^{M-N} =$$

$$= V_0^{M-N} \left(1 - \rho \frac{V}{M-N}\right)^{M-N} =$$

TODAY'S TOPICS 26/10/2026

- IMPLEMENTATION OF MC-NPT
- GIBBS FREE ENERGY
- HISTOGRAM REWEIGHTING WITH AN EXAMPLE OF APPLICATION
- ORIENTATIONAL MOVES OF UNIAXIAL PARTICLES
- GENERIC RIGID BODIES

$M \gg N$ I.E. $M-N \to +\infty$ $\lim_{M-N\to+\infty}\left(1-\frac{\beta P V}{M-N}\right)^{M-N} = e^{-\beta P V}$

$\lim_{x\to+\infty}\left(1+\frac{k}{x}\right)^{x} = e^{kx}$

$= V_0^{M-N} e^{-\beta P \overbrace{V}^{=-\beta P V}} = V_0^{M-N} e^{-\beta P V}$

$\beta P = \rho$ HEAT BATH HENCE:

$\rho(V, \vec{s}_1 \dots \vec{s}_N) \propto \overbrace{e^{N \ln V}}^{V^N} e^{-\beta P V} e^{-\beta U_N}$

STATE OF THE SYSTEM IS GIVEN BY $\{V, \vec{s}_1 \dots \vec{s}_N\}$ HENCE WE NEED TO UPDATE BOTH V AND $\{\vec{s}_1 \dots \vec{s}_N\}$, I.E. WE NEED TWO TYPES OF TRIAL MOVES: PARTICLE MOVE (UPDATING RANDOMLY A PARTICLE) AND A VOLUME MOVE

1) PARTICLE MOVE:

CHOOSE RANDOMLY A PARTICLE AND UPDATE ITS POSITION $\vec{s}_i$

NOW NOTE THAT IF WE UPDATE PARTICLE i POSITION AS IN A NVT MC, I.E.

$\vec{r}_i' = \vec{r}_i + \Delta\vec{r}$

THIS IS EQUIVALENT TO DO:

$V^{1/3}\vec{s}_i' = V^{1/3}\vec{s}_i + \Delta\vec{r}$ I.E.

$\vec{s}_i' = \vec{s}_i + V^{-1/3}\Delta\vec{r}$

I.E. WE ARE UPDATING $\vec{s}_i$ AS REQUESTED

THEN ACCEPT THE TRIAL MOVE AS FOLLOWS:

$$acc(o \to n) = \min\left\{1, \frac{e^{N \ln V} e^{-\beta P V} e^{-\beta U_N(V, \vec{s}_1 \dots \vec{s}_i' \dots \vec{s}_N)}}{e^{N \ln V} e^{-\beta P V} e^{-\beta U_N(V, \vec{s}_1 \dots \vec{s}_N)}}\right\}$$

$$= \min\left\{1, e^{-\beta \Delta U_N}\right\}$$

NOT THAT THIS MOVE IS EQUIVALENT TO THAT OF A NVT MC

2) VOLUME MOVE $o \quad n$

$\vec{r}_i = V^{1/3} \vec{s}_i$

$V \to V'$

$\vec{r}_i' = (V')^{1/3} \vec{s}_i$

$\vec{r}_i' = \vec{r}_i \left(\frac{V'}{V}\right)^{1/3}$

VOLUME MOVE

$V' = V + \delta V \, \eta$

$\eta \in (-1, 1)$ ← RANDOM

AND $acc(o \to n) = \min\left\{1, \frac{\rho(V', \vec{s}_1 \dots \vec{s}_N)}{\rho(V, \vec{s}_1 \dots \vec{s}_N)}\right\} =$

$\frac{\rho(n)}{\rho(o)} =$

$$= \min\left\{1, \frac{e^{-\beta U_N(V') - \beta P V' + N \ln V'}}{e^{-\beta U_N(V) - \beta P V + N \ln V}}\right\}$$

IF $G \equiv U_N + PV - k_B T N \ln V$

$$\Delta G = P \Delta V + \Delta U_N - N k_B T \ln\left(\frac{V(n)}{V(o)}\right)$$

$\Delta V = V' - V$

$\Delta U_N = U_N(V') - U_N(V)$

ACCEPTANCE PROBABILITY NPT:

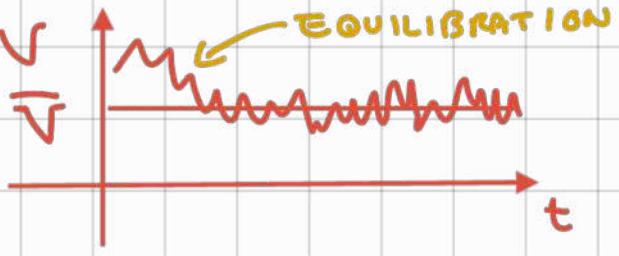
$$acc(o \to n) = \min\left\{1, e^{-\beta \Delta G}\right\}$$

NOTE THAT:

1) δV MUST BE OPTIMIZED TO HAVE AN ACCEPTANCE RATE OF VOLUME MOVES AROUND 30% – 50% (AS δ)

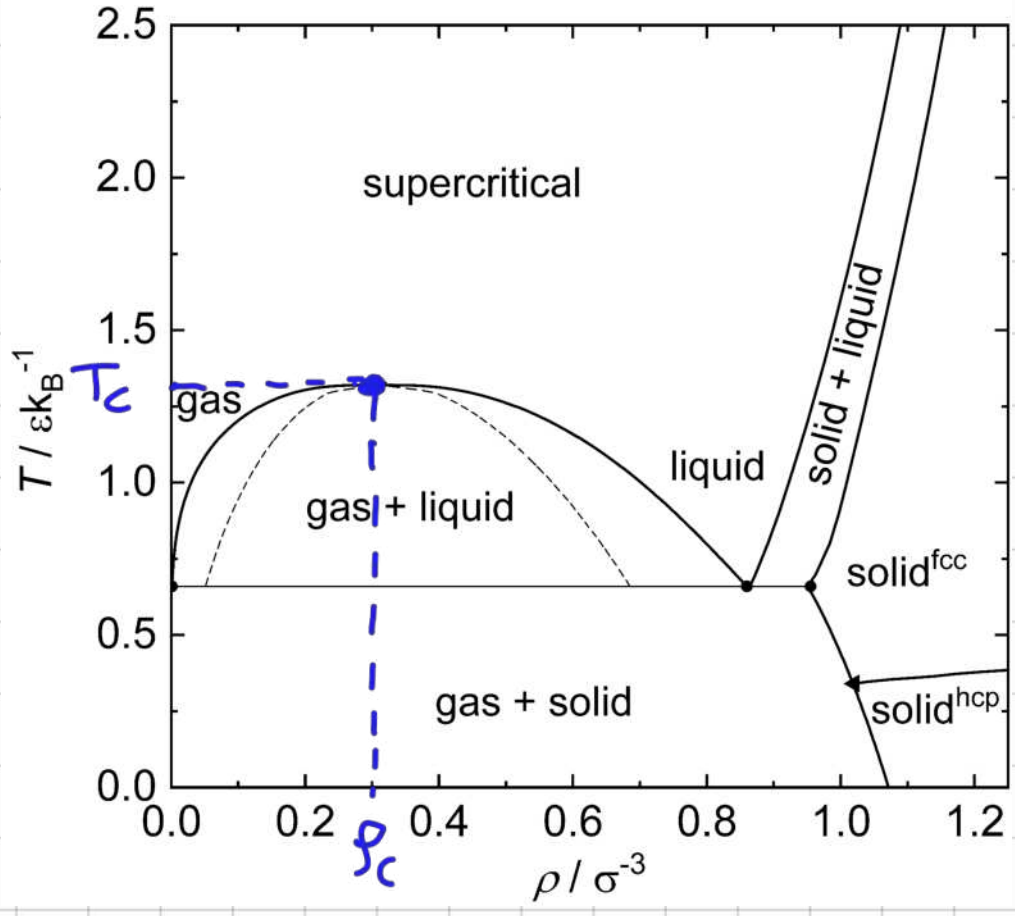
2) THE TWO TYPES OF MOVE SELECTED RANDOMLY WHERE THE PROBABILITY OF SELECTING A VOLUME MOVE SHOULD BE EQUAL TO $1/N$ (BECAUSE EACH BOX MOVE INVOLVES N PARTICLES)

3) MC NPT

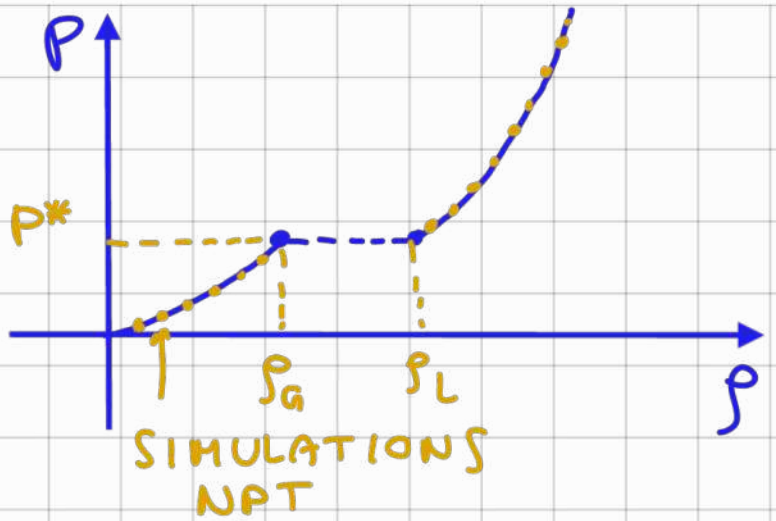


VOLUME PERFORMS A RANDOM WALK (FLUCTUATES)

LENNARD-JONES PHASE DIAGRAM



FOR $T < T_c$



PROBABILITY OF A GIVEN CONFIGURATION:

$$\rho(V, \vec{s}_1 \dots \vec{s}_N) = \frac{e^{-\beta U_N} e^{-\beta P V} V^N}{\int e^{-\beta U_N} e^{-\beta P V} V^N \, dV \, d^{3N}s}$$

PROBABILITY OF VOLUME V:

$$\rho(V) = \int \frac{e^{-\beta U_N} e^{-\beta P V} V^N}{\int e^{-\beta U_N} e^{-\beta P V} V^N \, dV \, d^{3N}s} d^N \vec{s} =$$

$$= \frac{1}{Z_{NPT}} e^{-\beta P V} \overbrace{\int e^{-\beta U_N} V^N d^N \vec{s}}^{Z(T,V)} = \frac{e^{-\beta P V} Z(T,V)}{Z_{NPT}}$$

WHERE $Z(T,V) = \int e^{-\beta U_N} V^N d^N \vec{s}$ AND

$$Z_{NPT} = \int e^{-\beta U_N} e^{-\beta P V} V^N \, dV \, d^{3N}s$$

GIBBS FREE ENERGY

$$Q_{NPT} \equiv \frac{Z_{NPT}}{\Lambda^{3N} N!}$$

← WE ALSO CONSIDER THE INTEGRAL OVER MOMENTA AND THE CORRECT BOLTZMANN COUNTING

$$-k_B T \ln(Q_{NPT}) = -k_B T \ln\left(\int e^{-\beta U_N} e^{-\beta P V} V^N \frac{dV \, d^{3N}s}{\Lambda^{3N} N!} \right)$$

CONTRIBUTE TO INTEGRAL WILL COME MOSTLY FROM $\bar{V} = \langle V \rangle$ HENCE WE CAN WRITE:

$$= -k_B T \ln\left(\int e^{-\beta U_N} e^{-\beta P \bar{V}} \bar{V}^N \frac{d^{3N}s}{\Lambda^{3N} N!} \right) =$$

$$= -k_B T \ln\left(e^{-\beta P \bar{V}} \int e^{\beta U_N} \bar{V}^N \frac{d^N s}{\lambda^{3N} N!} \right)$$

BUT $$\int e^{-\beta U_N} \bar{V}^N \frac{d^N s}{\lambda^{3N} N!} = Q_N(T, \bar{V})$$

↑ CANONICAL PARTITION FUNCTION

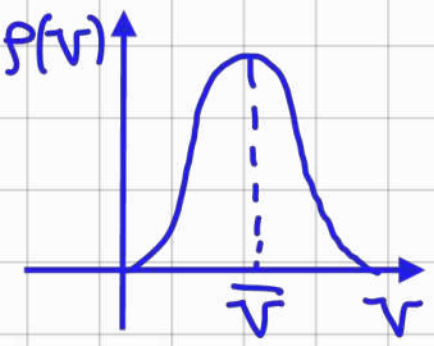
HENCE

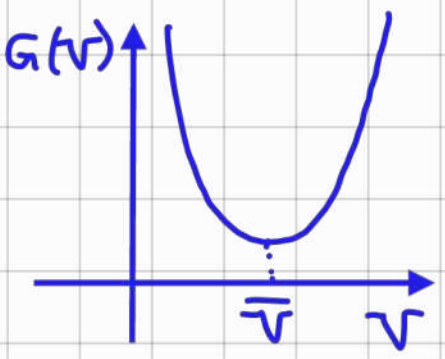
$$= P\bar{V} - k_B T \ln Q_N(T, \bar{V}) = P\bar{V} + F(T, \bar{V}) =$$

$$= G(T, P) \leftarrow \text{GIBBS FREE ENERGY}$$

$$G(V) \equiv -k_B T \ln \rho(V) + K \leftarrow \text{CONSTANT WHICH IS INDEPENDENT OF } V$$

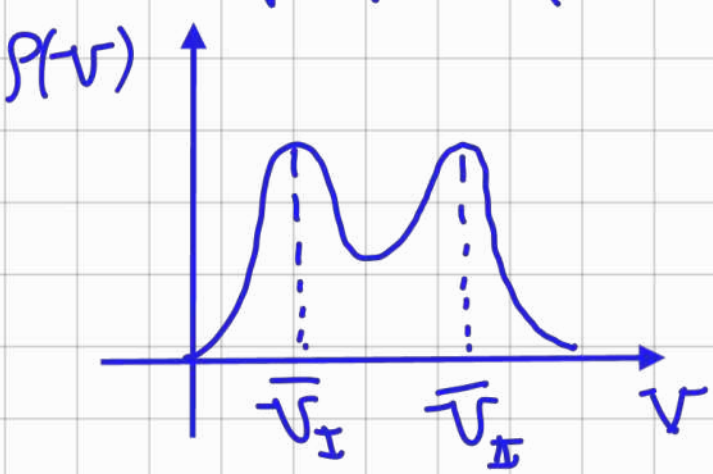
ONE PHASE

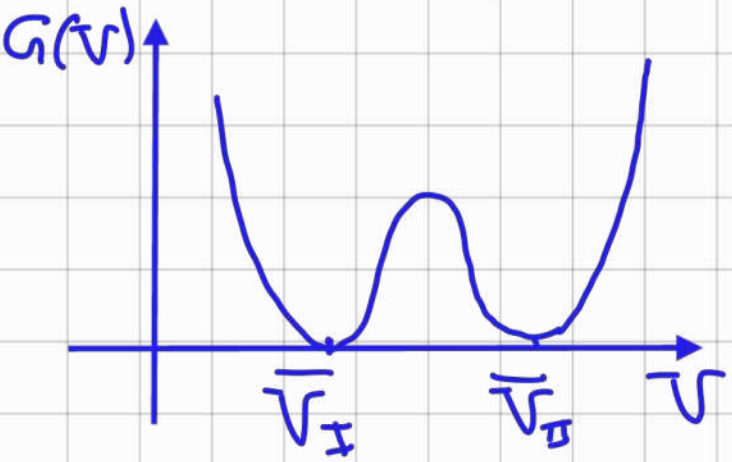




TWO PHASE COEXISTENCE (I AND II PHASES)

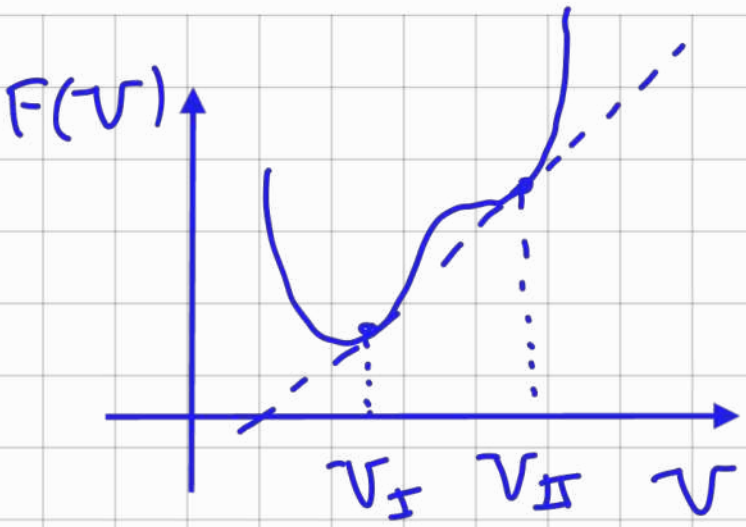
$P = P^* = \{\text{COEXISTENCE PRESSURE}\}$





$$F(V) = G(V) - PV$$

↑ INTENDED AS $\bar{V} = \langle V \rangle$

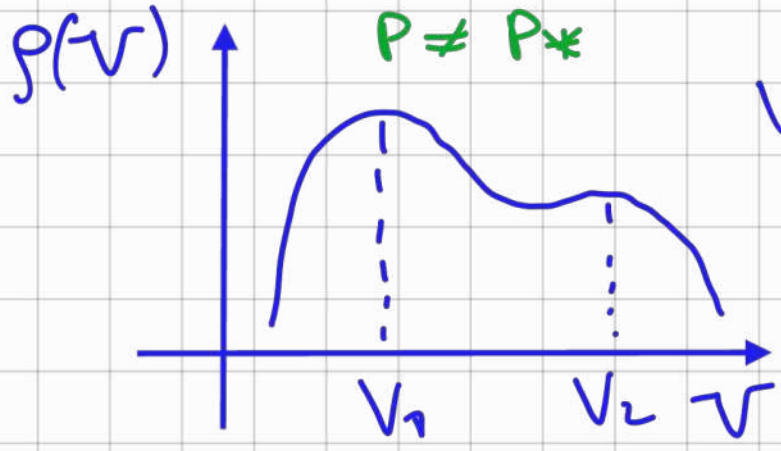


$$\left.\frac{\partial F}{\partial V}\right|_{V=V_I} = -P^*$$

$$\left.\frac{\partial F}{\partial V}\right|_{V=V_{II}} = -P^*$$

↑ COMMON TANGENT

IN GENERAL $P \neq P^*$ HENCE WE CAN NOT BE EXACTLY AT COEXISTENCE, I.E.



$V_1 \neq V_I$

$V_2 \neq V_{II}$

IF $P \neq P^*$ POSITIONS OF THE PEAKS ARE NOT THOSE OF COEXISTENCE

HISTOGRAM REWEIGHTING

$$p(V, P_1) = \frac{e^{-\beta P_1 V} \, \mathcal{Z}(T,V)}{\mathcal{Z}_{NPT}(T, P_1)}$$

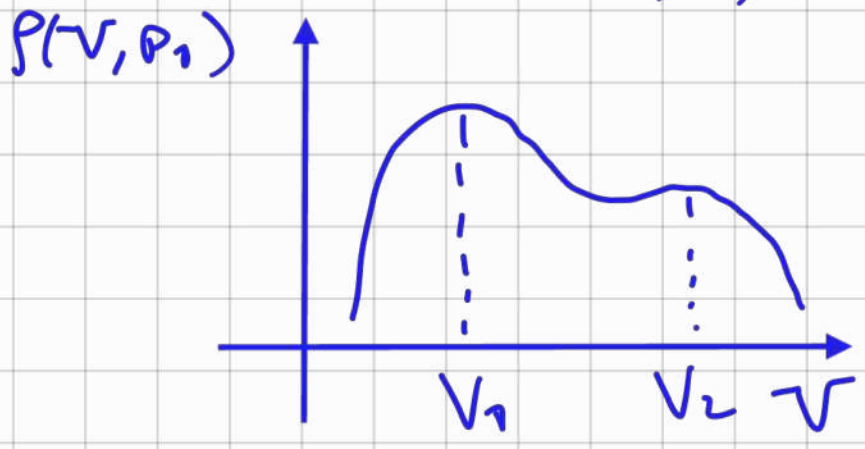
← FROM A SIMULATION AT $P = P_1 \neq P^*$

$$p(V, P_2) = \frac{e^{-\beta P_2 V} \, \mathcal{Z}(T,V)}{\mathcal{Z}_{NPT}(T, P_2)}$$

$$\frac{p(V, P_2)}{p(V, P_1)} = \frac{e^{-\beta P_2 V} \, \cancel{\mathcal{Z}(T,V)} \, \mathcal{Z}_{NPT}(T, P_1)}{e^{-\beta P_1 V} \, \cancel{\mathcal{Z}(T,V)} \, \mathcal{Z}_{NPT}(T, P_2)} =$$

$$\rho(V,P_2) = \rho(V,P_1)\cdot e^{-\beta(P_2-P_1)V}\, K \qquad (1)$$

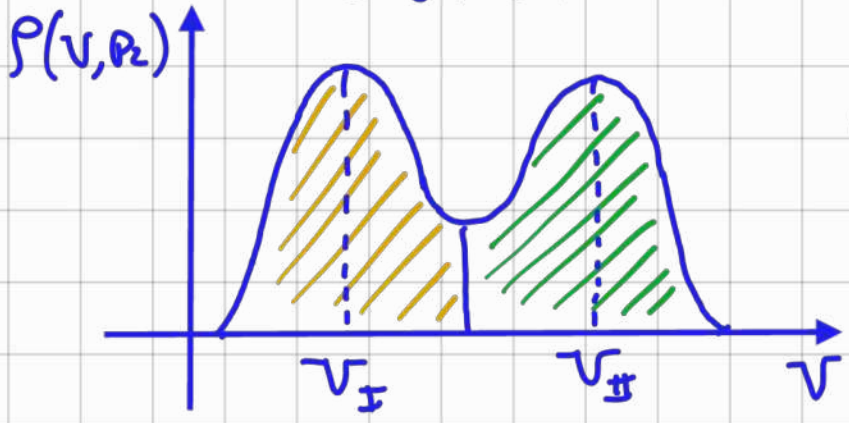
WHERE $K = \dfrac{Z_{NPT}(T,P_1)}{Z_{NPT}(T,P_2)}$ ← WHICH IS INDEPENDENT OF V (NORMALIZATION)



$P_1 > P^*$

BY USING EQ. (1) WE CAN SEARCH FOR P_2 SUCH THAT THE TWO FOLLOWING AREAS (/// AND ////) ARE EQUAL (I.E. THE TWO PHASES ARE EQUALLY LIKELY)

HISTOGRAM REWEIGHTING



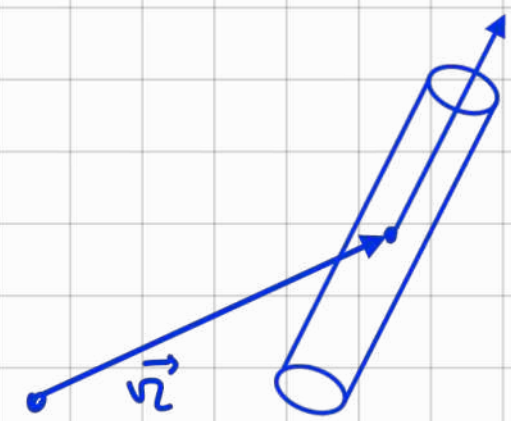
$$\rho_I = \frac{N}{V_I} \qquad \rho_{II} = \frac{N}{V_{II}}$$

IF WE ARE NOT AT COEXISTENCE WE CAN EMPLOY HISTOGRAM REWEIGHTING, ANYWAY WE HAVE TO BE CLOSE ENOUGH ($P_1 \approx P^*$) SO THAT $\rho(V_{II},P_1)$ IS NOT NEGLIGIBLE (WITHIN SIMULATION TIMES)

SHOW histogram_reweighting.nb (MATHEMATICA)

ORIENTATIONAL MOVES

UNIAXIAL PARTICLE
(E.G. CYLINDERS)



$\vec{u}$ (ORIENTATION OF PARTICLE)

(PARTICLE POSITION)

SYSTEM STATE

$(\vec{r}_1, \vec{u}_1 \dots \vec{r}_N \vec{u}_N)$

CHOOSE RANDOMLY WHETHER DOING A TRANSLATIONAL OR A ROTATIONAL MOVE

TRANSLATIONAL MOVE IS LIKE WE ALREADY DISCUSSED

ROTATIONAL TRIAL MOVE ($o \to n$)

$$\vec{u}_o \to \vec{u}_n = \frac{\vec{u}_o + \gamma \Delta \vec{u}}{\| \vec{u}_o + \gamma \Delta \vec{u} \|}$$

WHERE $\Delta \vec{u}$ IS A RANDOM UNIT VECTOR (MARSAGLIA ALGORITHM)

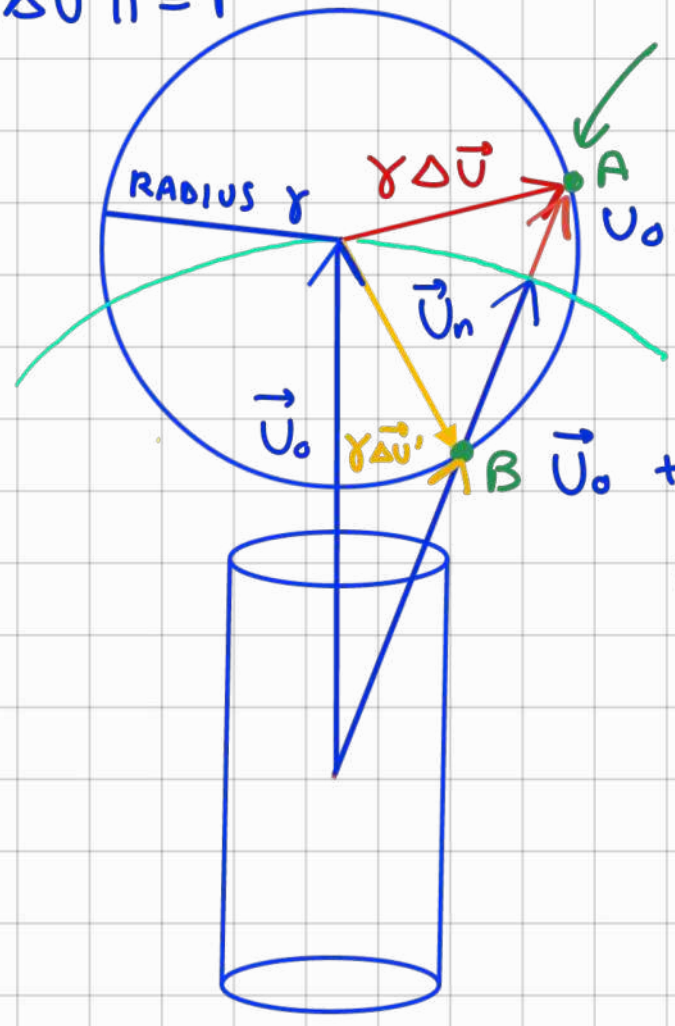
γ MUST BE OPTIMIZED AS δ ACCEPTANCE DOES NOT CHANGE

THIS MOVE IS SYMMETRIC, I.E.

$$\alpha(0 \to n) = \alpha(n \to 0)$$

$$\alpha(0 \to n)$$

$\|\Delta\vec{v}\| = \|\Delta\vec{v}'\| = 1$



POINTS A AND B: INTERSECTIONS OF DIRECTION $\vec{v}_n$ WITH A SPHERE OF RADIUS γ

$$v_n = \frac{\vec{v}_0 + \gamma\Delta\vec{v}}{\|\vec{v}_0 + \gamma\Delta\vec{v}\|} = \frac{\vec{v}_0 + \gamma\Delta\vec{v}'}{\|\vec{v}_0 + \gamma\Delta\vec{v}'\|}$$

I.E.

$\Delta\vec{v}$ AND $\Delta\vec{v}'$ GIVE THE SAME $\vec{v}_n$

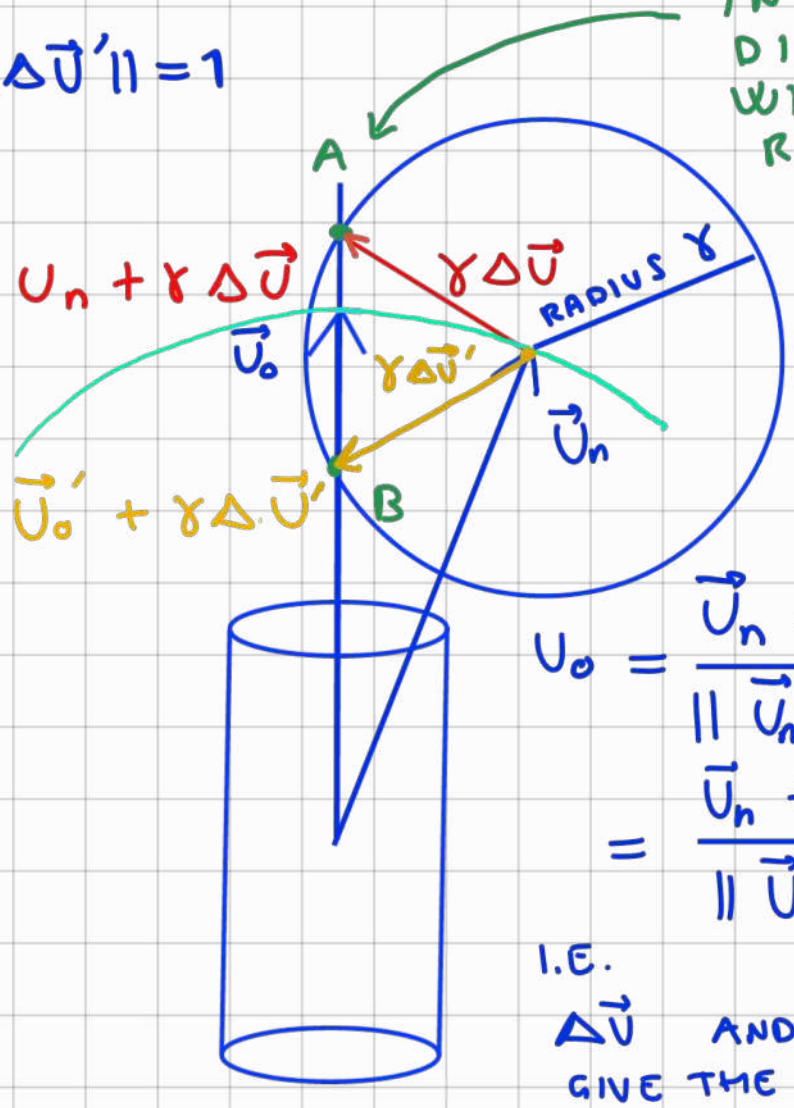
IF YOU DO SAME CONSTRUCTION FROM $\vec{v}_n$ YOU WILL HAVE THAT PROBABILITY IS THE SAME (CHANCE OF GENERATING TWO ORIENTATIONS ON UNIT SPHERE)

INDEED:

$2(n \to o)$

$\|\Delta\vec{v}\| = \|\Delta\vec{v}'\| = 1$

POINTS A AND B. INTERSECTIONS OF DIRECTION $\vec{v}_o$ WITH A SPHERE OF RADIUS γ



$$v_o = \frac{\vec{v}_n + \gamma\Delta\vec{v}}{\|\vec{v}_n + \gamma\Delta\vec{v}\|} =$$

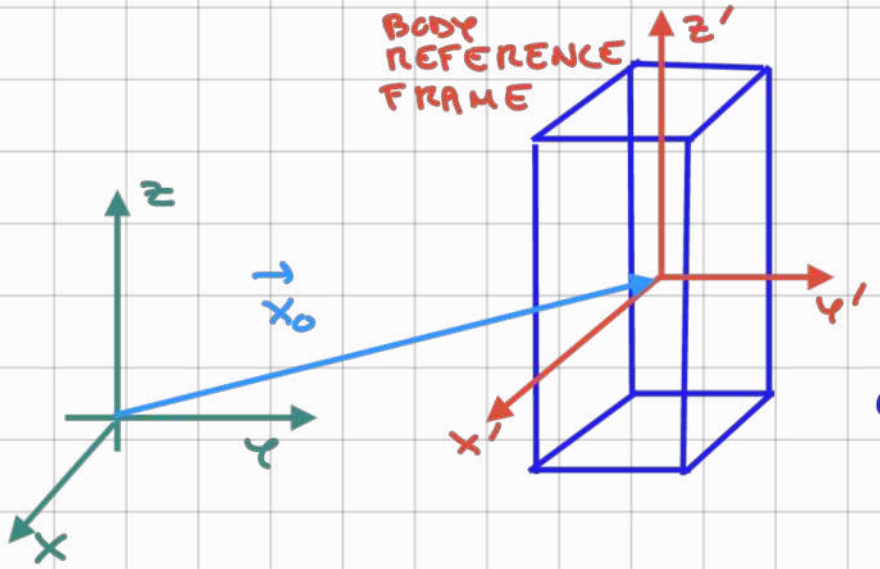
$$= \frac{\vec{v}_n + \gamma\Delta\vec{v}'}{\|\vec{v}_n + \gamma\Delta\vec{v}'\|}$$

I.E.

$\Delta\vec{v}$ AND $\Delta\vec{v}'$ GIVE THE SAME $\vec{v}_o$

HENCE $2(o \to n) = 2(n \to o)$, I.E. PROBABILITY IS THE SAME BEING THE CHANCE OF GENERATING TWO ORIENTATIONS ON UNIT SPHERE

NOT UNIAXIAL PARTICLES



SYSTEM STATE

$(\vec{r}_1, \Omega_1 \ldots \vec{r}_N \Omega_N)$

E.g. $\Omega_i = (\theta_i, \psi_i, \phi_i)$ EULER ANGLES

ALTERNATIVE TO EULER ANGLES

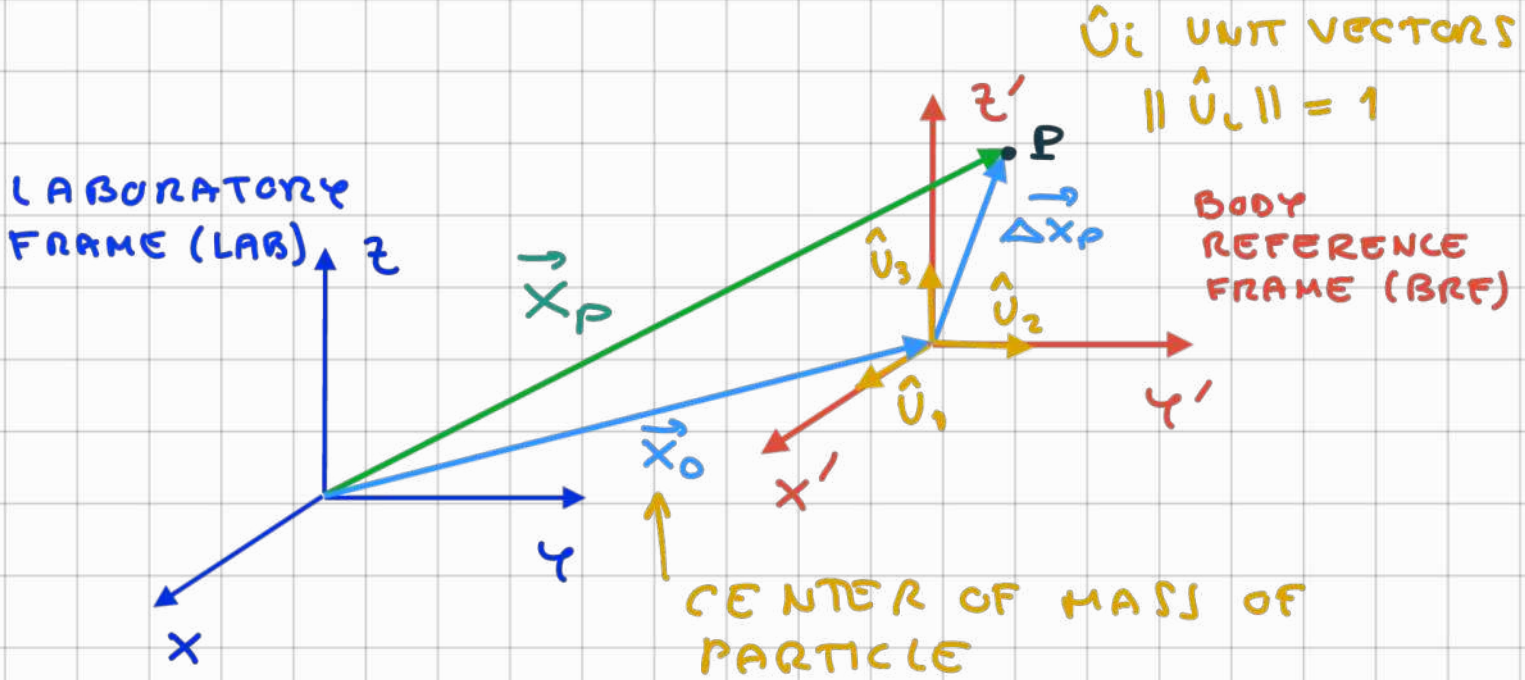
ROTATION MATRIX R SUCH THAT

$$\vec{x}' = R(\vec{x} - \vec{x}_0)$$

$\vec{x}'$: BODY REFERENCE FRAME

$\vec{x}$: LABORATORY FRAME

CLOSER LOOK TO ROTATION MATRIX



$x_0 = (x, y, z)$ LAB

$x_0' = (x', y', z')$ BRF

VECTORS WHOSE COORDINATES ARE EXPRESSED IN LAB FRAME

$$\vec{x}_P = \vec{x}_0 + (\vec{x}_P - \vec{x}_0) = \vec{x}_0 + \Delta\vec{x}_P$$

$$\Delta\vec{x}_P = \vec{x}'\,\hat{u}_1 + \vec{y}'\,\hat{u}_2 + z'\,\hat{u}_3$$

← BRF COORDINATES OF P

I.E.

$\hat{u}_1$ $\hat{u}_2$ $\hat{u}_3$ ← INTENDED AS COLUMN VECTORS

$$\Delta\vec{x}_P = \begin{pmatrix} u_{1x} & u_{2x} & u_{3x} \\ u_{1y} & u_{2y} & u_{3y} \\ u_{1z} & u_{2z} & u_{3z} \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} =$$

COLUMN VECTORS

$$= \begin{pmatrix} \hat{u}_1 & \hat{u}_2 & \hat{u}_3 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$$

IF WE DEFINE

$$R \equiv \begin{pmatrix} u_{1x} & u_{1y} & u_{1z} \\ u_{2x} & u_{2y} & u_{2z} \\ u_{3x} & u_{3y} & u_{3z} \end{pmatrix}$$

← EACH ROW IS A UNIT VECTOR

THEN

$${}^tR = \begin{pmatrix} u_{1x} & u_{2x} & u_{3x} \\ u_{1y} & u_{2y} & u_{3y} \\ u_{1z} & u_{2z} & u_{3z} \end{pmatrix}$$

TRANSPOSE OF R

SO THAT

$$\Delta\vec{x}_P = {}^tR\,\vec{x}_P'$$

NOTE THAT:

$${}^tR\,R = \mathbb{1}$$

HENCE $${}^tR = R^{-1}$$

I. E.

LAB → $\vec{x}_P = \vec{x}_0 + {}^tR\,\vec{x}_P'$ ← BRF

$$\vec{x}_P' = R\,(\vec{x}_P - \vec{x}_0)$$

GIVEN A POINT IN THE BODY REFERENCE FRAME ITS COORDINATES IN THE LAB REFERENCE FRAME CAN BE OBTAINED THROUGH THE ROTATION MATRIX

IF WE ROTATE A PARTICLE BY AN ANGLE $\Delta\theta$ AROUND AN AXIS $\hat{U}_R = (U_x\ U_y\ U_z)$

WE HAVE

$$R' = R\left(I + M(\Delta\theta)\right) \qquad (1)$$

(I: IDENTITY MATRIX; $M(\Delta\theta)$: A MATRIX DEFINED IN THE FOLLOWING)

WHERE

$$M(\Delta\theta) = -\sin(\Delta\theta)\,\Omega + (1-\cos(\Delta\theta))\,\Omega^2$$

WITH

$$\Omega = \begin{pmatrix} 0 & -U_z & U_y \\ U_z & 0 & -U_x \\ -U_y & U_x & 0 \end{pmatrix}$$

(COORDINATES OF ROTATION AXIS IN THE LAB FRAME)

HENCE A ROTATIONAL TRIAL MOVE $\alpha(o \to n)$ OF PARTICLE i

$$(\vec{r}_i, R_i) \longrightarrow (\vec{r}_i, \vec{R}_i')$$

OLD(o) NEW(n)

R_i IS UPDATED ACCORDING TO EQ. (1) WHERE $\vec{U}_R$ IS CHOSEN RANDOMLY ON THE UNIT SPHERE AND $\Delta\theta$ IS RANDOM

I.E. $\Delta\theta = \Delta\theta_{MAX}\,\xi_\theta$ WITH $\xi_\theta \in [-1, 1]$

WHERE $\Delta\theta_{MAX}$ MUST BE OPTIMIZED IN ORDER TO HAVE AN ACCEPTANCE RATIO FOR ROTATIONAL MOVE AROUND 30-50%

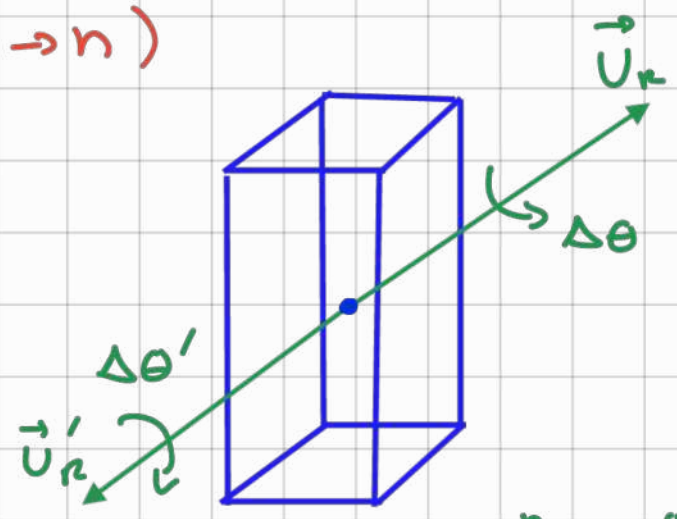
$\alpha(o \to n) = \alpha(n \to o)$

NOTE THAT THIS MOVE IS SYMMETRIC SINCE TO GO BACK WE HAVE TO CHOOSE SAME ORIENTATION $\vec{U}_R$ AND A ROTATION ANGLE $-\Delta\theta$

THIS ORIENTATIONAL MOVE IS SYMMETRIC

$d(0 \to n) = d(n \to 0)$

$d(0 \to n)$



I: $\vec{U}_R$ $\Delta\theta$ AND

II: $\vec{U}'_R = -\vec{U}_R$ $\Delta\theta' = -\Delta\theta$

GIVE SOME NEW ORIENTATION

REMEMBER THAT $R^n = R^0(I + M(\Delta\theta))$ WITH

$M(\Delta\theta) = -\sin(\Delta\theta)\,\Omega + (1 - \cos(\Delta\theta))\,\Omega^2$

IF $\vec{U} \to -\vec{U}$ AND $\Delta\theta \to -\Delta\theta$

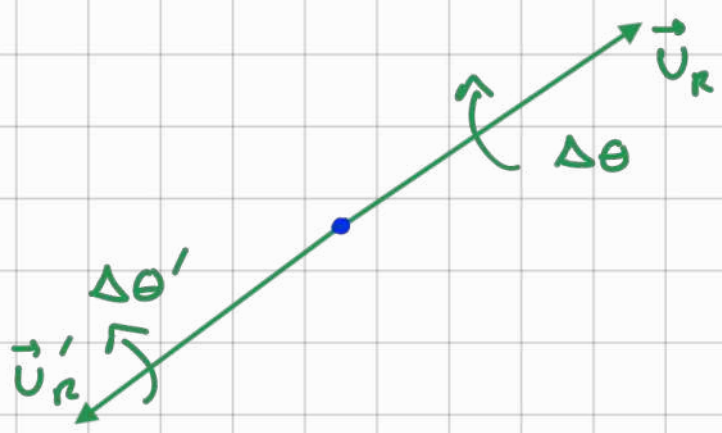
$\sin(\Delta\theta) \to -\sin(\Delta\theta)$

$\cos(\Delta\theta) \to \cos(-\Delta\theta) = \cos(\Delta\theta)$

$\Omega \to -\Omega$

HENCE M IS IDENTICAL FOR ROTATIONS I AND II

$d(n \to 0)$



ROTATIONS

$\vec{U}_R$ $-\Delta\theta$ AND

$\vec{U}'_R = -\vec{U}_R$ $\Delta\theta' = \Delta\theta$

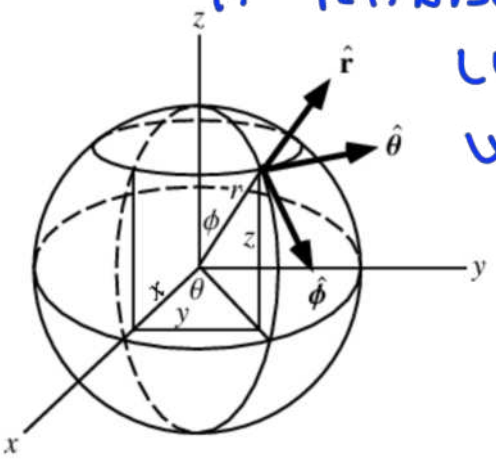
GIVE SAME OLD ORIENTATION

HENCE $d(0 \to n) = d(n \to 0)$

Spherical Coordinates

DOWNLOAD Wolfram Notebook EXPLORE THIS TOPIC IN The MathWorld Classroom

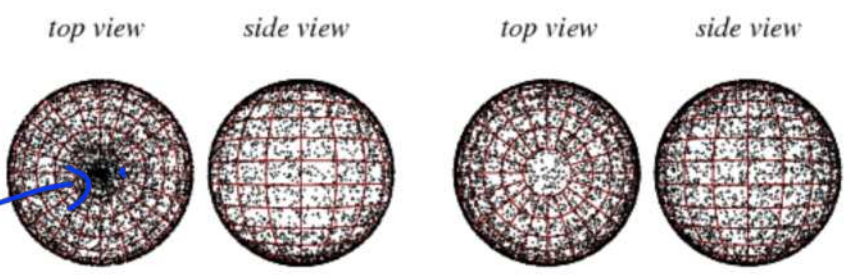
FOR GENERATING A RANDOM UNIT VECTOR LEAD TO WRONG RESULTS



$$d\Omega = \sin\theta \, d\theta \, d\varphi$$

Sphere Point Picking

DOWNLOAD Wolfram Notebook



incorrectly distributed points *correctly distributed points*

To pick a random point on the surface of a unit sphere, it is incorrect to select spherical coordinates θ and ϕ from uniform distributions $\theta \in [0, 2\pi)$ and $\phi \in [0, \pi]$, since the area element $d\Omega = \sin\phi \, d\theta \, d\phi$ is a function of ϕ, and hence points picked in this way will be "bunched" near the poles (left figure above).

HIGHER PROBABILITY AT POLES

SEE MARSAGLIA ALGORITH FOR A CORRECT WAY OF RANDOM UNIT VECTORS

ANOTHER POSSIBILITY: QUATERNIONS

REPRESENTS PARTICLE ORIENTATION

$\vec{q} = (q_0, q_1, q_2, q_3) \qquad \|\vec{q}\| = 1$

INDEED EACH CAN BE MAPPED TO A ROTATION MATRIX AND VICE-VERSA

$R \rightarrow \vec{q}$

$$\mathbf{q}_{\mathrm{IB}} = \begin{bmatrix} \cos(\psi/2) \\ 0 \\ 0 \\ \sin(\psi/2) \end{bmatrix} \begin{bmatrix} \cos(\theta/2) \\ 0 \\ \sin(\theta/2) \\ 0 \end{bmatrix} \begin{bmatrix} \cos(\phi/2) \\ \sin(\phi/2) \\ 0 \\ 0 \end{bmatrix}$$

EULER ANGLES

$$= \begin{bmatrix} \cos(\phi/2)\cos(\theta/2)\cos(\psi/2) + \sin(\phi/2)\sin(\theta/2)\sin(\psi/2) \\ \sin(\phi/2)\cos(\theta/2)\cos(\psi/2) - \cos(\phi/2)\sin(\theta/2)\sin(\psi/2) \\ \cos(\phi/2)\sin(\theta/2)\cos(\psi/2) + \sin(\phi/2)\cos(\theta/2)\sin(\psi/2) \\ \cos(\phi/2)\cos(\theta/2)\sin(\psi/2) - \sin(\phi/2)\sin(\theta/2)\cos(\psi/2) \end{bmatrix}$$

EULER ANGLES CAN BE OBTAINED FROM THE ROTATION MATRIX

$\vec{q} \rightarrow R$

$$R = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1 q_2 - q_0 q_3) & 2(q_0 q_2 + q_1 q_3) \\ 2(q_1 q_2 + q_0 q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_0 q_1 + q_2 q_3) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix}$$

STATE OF THE SYSTEM

$(\vec{r}_1, \vec{q}_1 \ldots \vec{r}_N, \vec{q}_N) \qquad \|\vec{q}_i\| = 1$

THE ROTATIONAL TRIAL MOVE $\alpha(o \to n)$ CONSISTS IN DOING:

$$\vec{q}_n = \frac{\vec{q}_o + \gamma \Delta\vec{q}}{\| \vec{q}_o + \gamma \Delta\vec{q} \|}$$

γ MUST BE OPTIMIZED LIKE δ FOR TRANSLATIONS

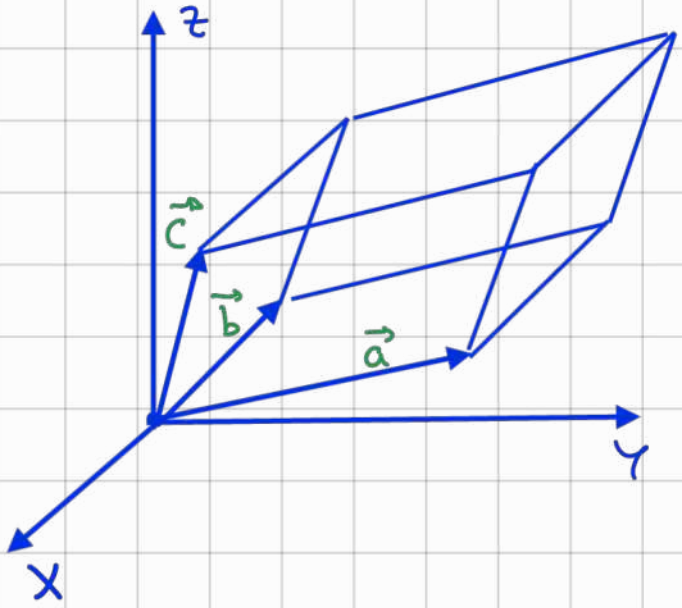
WHERE $\Delta\vec{q}$ IS A UNIT RANDOM VECTOR ON THE 4D UNIT SPHERE (AGAIN BY USING MARSAGLIA ALGORITHM)

$$\gamma = \gamma_{MAX}\, \xi_\gamma \qquad \xi_\gamma \in [-1, 1]$$

AND γ_{MAX} MUST BE OPTIMIZED TO HAVE 30-50% ACCEPTANCE RATE OF ORIENTATIONAL MOVES

PARRINELLO-RAHMAN

ALLOW FOR CHANGES OF BOX SHAPE



$\vec{a} = (a_x, a_y, a_z)$

$\vec{b} = (b_x, b_y, b_z)$

$\vec{c} = (c_x, c_y, c_z)$

↑ EDGES OF SIMULATION BOX (NOT ORTHOGONAL)

$$Q = \underbrace{\frac{(V_0 - V)^{M-N}}{\Lambda^{3(M-N)}\,(M-N)!}}_{\text{HEAT BATH}} \underbrace{\int \frac{e^{-\beta U_N}}{\Lambda^{3N}\,N!}\, d^N\vec{r}}_{\substack{\text{OUR SYSTEM WITH NON ORTHOGONAL EDGES}}}$$

REDUCED COORDINATES

$$\vec{r}_i = \begin{pmatrix} a_x & b_x & c_x \\ a_y & b_y & c_y \\ a_z & b_z & c_z \end{pmatrix} \vec{s}_i$$

$$\| \quad M$$

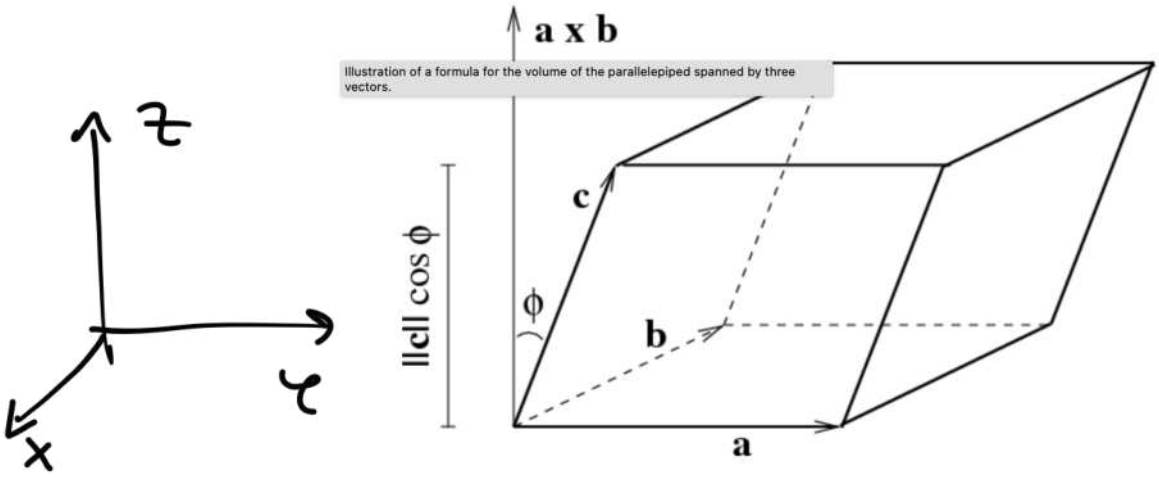
NOTE THAT IF $M = \begin{pmatrix} L & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & L \end{pmatrix}$

WE HAVE STANDARD NPT ENSEMBLE

$$Q = \frac{(V-V_0)^{M-N}}{\Lambda^{3(M-N)}(M-N)!} \int \frac{e^{-\beta U_N}}{\Lambda^{3N} N!} |\det M|^N d^N s$$

BUT

BOX VOLUME



This formula for the volume can be understood from the above figure. The volume of the parallelepiped is the area of the base times the height. From the geometric definition of the cross product, we know that its magnitude, $\|\mathbf{a} \times \mathbf{b}\|$, is the area of the parallelogram base, and that the direction of the vector $\mathbf{a} \times \mathbf{b}$ is perpendicular to the base. The height of the parallelepiped is the component of $\mathbf{c}$ in the direction normal to the base, i.e., in the direction of $\mathbf{a} \times \mathbf{b}$. Hence the height is $\|\mathbf{c}\| \, |\cos\phi|$, where ϕ is the angle between $\mathbf{c}$ and $\mathbf{a} \times \mathbf{b}$. (Why do we need the absolute value? If we swapped the locations of $\mathbf{a}$ and $\mathbf{b}$, in the above figure, then $\mathbf{a} \times \mathbf{b}$ would point downward. The angle ϕ would be larger than $\pi/2$, and $\cos\phi$ would be negative.)

$$V = |\vec{c} \cdot (\vec{a} \times \vec{b})| = \det \begin{pmatrix} a_x & b_x & c_x \\ a_y & b_y & c_y \\ a_z & b_z & c_z \end{pmatrix} = \det M$$

SO THAT

$$\rho(V, \vec{s}_1 \dots \vec{s}_N) \propto e^{-\beta U_N} e^{-\beta P V} V^N$$

AND

$$acc(o \to n) = \min\left\{1, \frac{\rho(V', s_1' \dots s_N')}{\rho(V, s_1 \dots s_N)}\right\} =$$

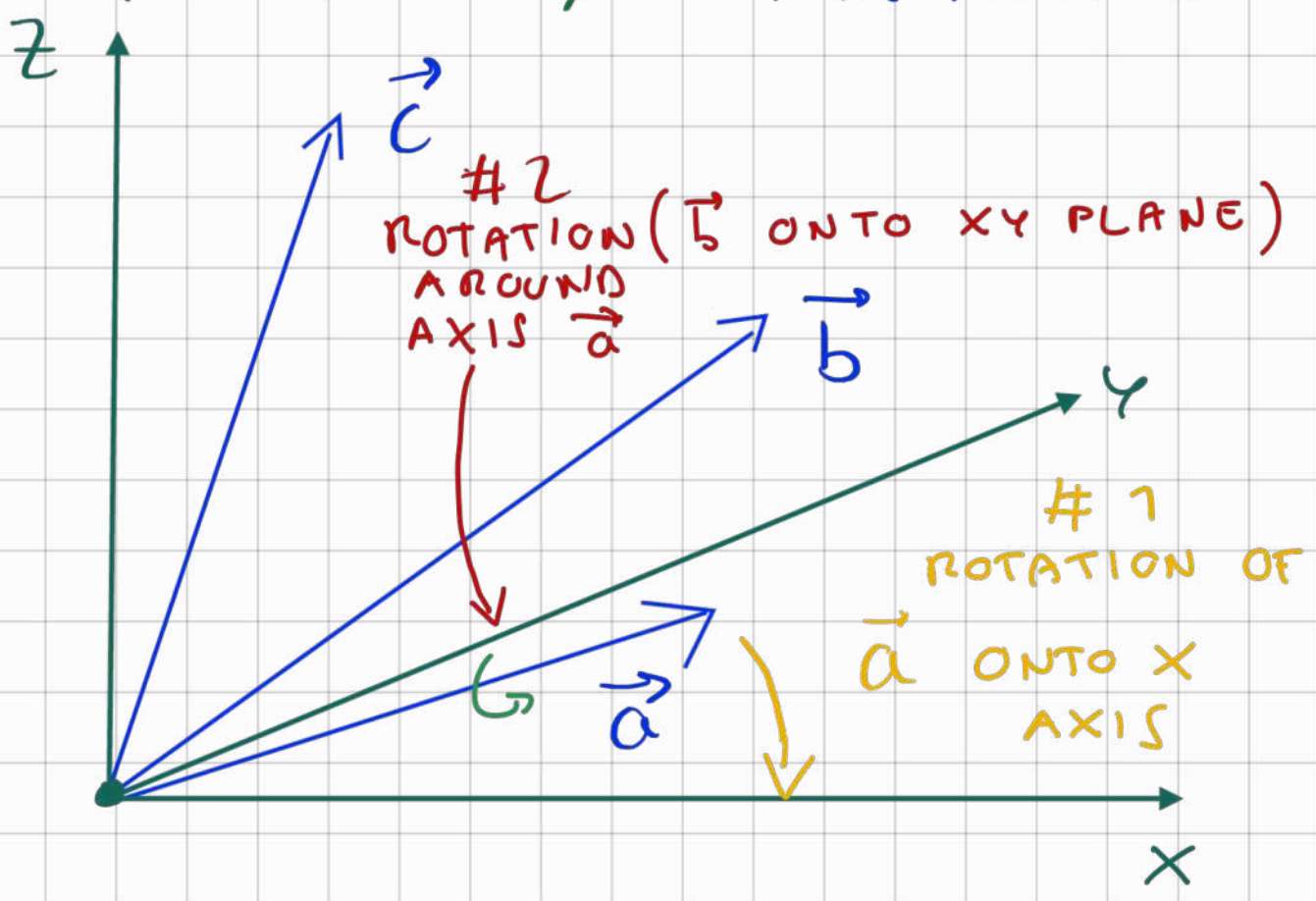
$$\min\{1, e^{-\beta \Delta G}\}$$

$$G = U + PV - k_B T N \ln U$$

IN GENERAL

$$M = \begin{pmatrix} a_x & b_x & c_x \\ 0 & b_y & c_y \\ 0 & 0 & c_z \end{pmatrix}$$

← BY DOING #1 AND #2 WE CAN REDUCE TO THIS MATRIX



XYZ IS THE LAB FRAME

SPECIAL CASES

1) $$M = \begin{pmatrix} \delta L_x & 0 & 0 \\ 0 & \delta L_y & 0 \\ 0 & 0 & \delta L_z \end{pmatrix}$$

ANISOTROPIC BOX (I.E. NOT CUBIC) BUT ISOTROPIC VOLUME MOVES

IN THIS CASE γ CHANGES RANDOMLY IN A VOLUME MOVE, I.E.

$\gamma' = \gamma + \Delta\gamma \qquad \Delta\gamma = \Delta\gamma_0 \, \eta$

$\eta \in [-1, 1]$ IS A STOCHASTIC VARIABLE

2) $M = \begin{pmatrix} L_x & 0 & 0 \\ 0 & L_y & 0 \\ 0 & 0 & L_z \end{pmatrix}$ ANISOTROPIC BOX WHERE THE 3 EDGE SIZE CAN CHANGE INDIPENDENTLY

IN A VOLUME MOVE A DIRECTION α IS SELECTED RANDOMLY AND

$L'_\alpha = L_\alpha + \Delta L$

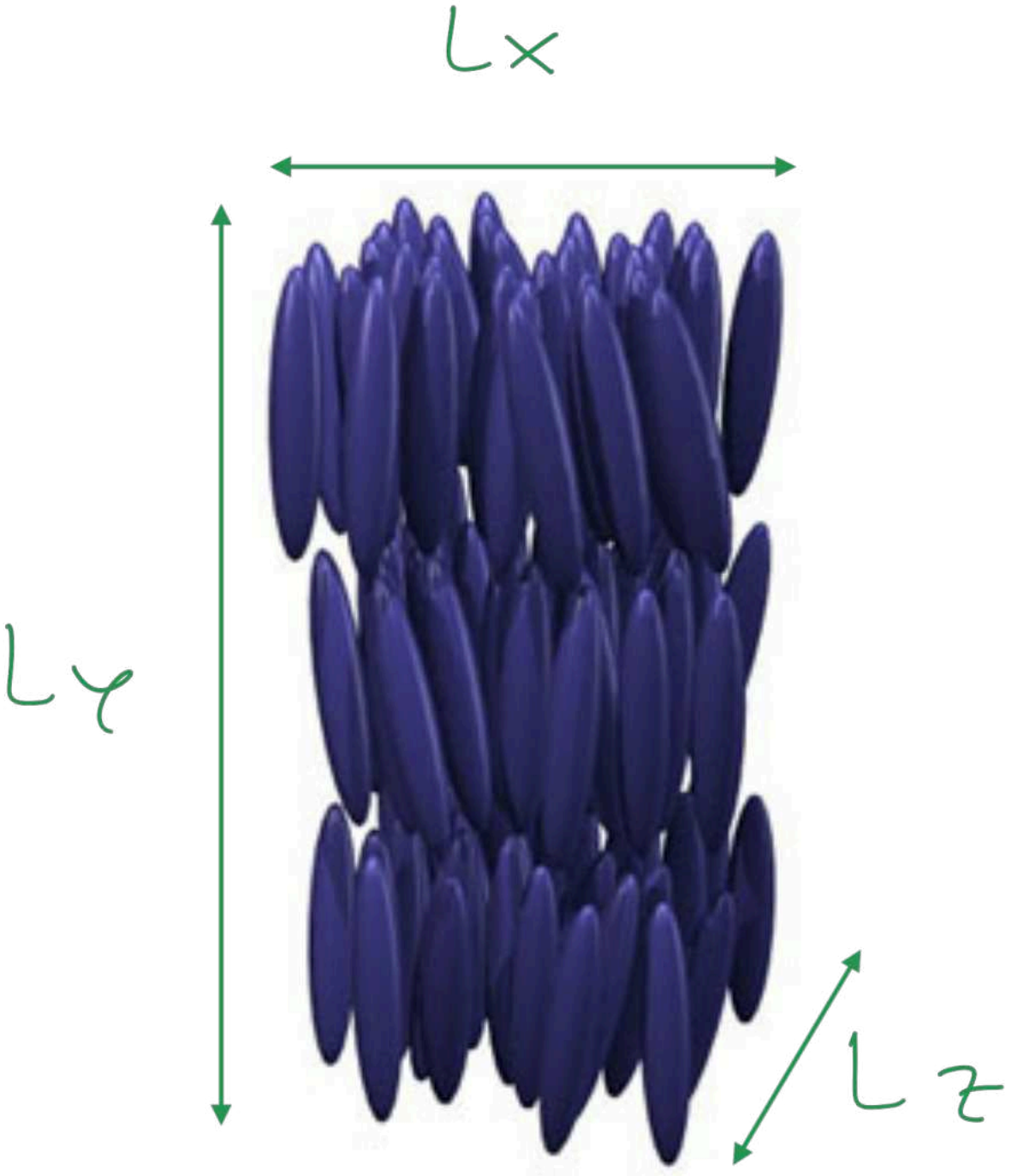
WHERE $\alpha \in \{x, y, z\}$

AND $\Delta L = \Delta L_0 \, \eta \qquad \eta \in [-1, 1]$ (RANDOM)

THIS TYPE OF MOVE CAN BE USEFUL FOR CRYSTALS AND LIQUID CRYSTALS

(SEE BELOW)

EXAMPLE: SMECTIC PHASE



$L_y \neq L_x, L_z$